



Semen Cuscutae-Fructus Lycii improves spermatogenic dysfunction by repairing the blood-testis barrier in rats according to in silico and in vitro methods

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ABSTRACT

Ethnopharmacological relevance: Semen Cuscutae and Fructus Lycii (SC-FL) is a commonly used herbal pair for male infertility treatment. Studies have found that the mechanism of SC-FL treatment may be related to repairing the blood-testis barrier (BTB). The application of network pharmacology can be used to explore the correlation between medicines and diseases and predict the potential pharmacological mechanisms of SC-FL.

Aim of the study: This study aimed to explore the specific effects and mechanisms of SC-FL in repairing the BTB and initially revealed the mechanism of Chinese medicine treating male infertility through network pharmacology and animal experiments.

Materials and methods: We searched databases using the network pharmacology method and performed mass spectrometry analysis. We analyzed and predicted the active ingredients, targets and key pathways of SC-FL in male infertility treatment. Then, we designed animal experiments to verify the results. Thirty-six Sprague-Dawley rats were randomly divided into the normal control group (NC group), spermatogenic dysfunction group (SD group) and SC-FL treatment group (SCFL group). Glucosides of Tripterygium wilfordii Hook. f. (GTW) (40 mg/kg/d) was administered for 4 weeks to generate a spermatogenic dysfunction model. The rats in the SCFL group were given the SC-FL suspension (6 g/kg/d) daily. After 4 weeks of treatment, we detected the sperm quality of each group of rats and observed the cell morphology. Western blotting and qRT-PCR were used to detect the expression of BTB-related proteins in testicular tissues.

Results: 213 chemical ingredients of SC and FL were retrieved from the TCMS database, and 54 effective chemical ingredients were obtained. Mass spectrometry analysis showed the above results were credible. Then, we identified 44 potential targets for the treatment of male infertility, and we plotted a network diagram of the interaction network between the core targets and a diagram of herbal medicine-active ingredient-target-disease interactions. The target genes were enriched according to biological functions, and 22 biological processes, 49 cellular components, 1487 molecular functions, and 122 signaling pathways were obtained. The results of the animal experiments showed that the sperm concentration and motility of the SCFL group were significantly improved compared with those of the SD group. Compared with those in the SD group, the structure and morphology of the Sertoli cells and seminiferous tubules of rats in the SCFL group improved, and the number of spermatogenic cells increased significantly. Western blotting and qRT-PCR results showed that compared with that in the SD group, the expression of p38 MAPK decreased significantly, and the expression of c-Jun, Occludin, ZO-1 and connexin 43 increased significantly in the SCFL group.

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Conclusion: We predicted that the active ingredients of SC-FL can treat male infertility by interacting with the core targets JUN, IL6, MAPK1, TP53, MYC, CCND1, AR, EGF, FOS, and MAPK8, and the possible mechanism is related to the MAPK signaling pathway. SC-FL can regulate the MAPK pathway and affect the expression of Occludin, ZO-1 and connexin 43 to repair damaged BTB and improve spermatogenic dysfunction induced by GTW, which may be one of the possible mechanisms.

1. Introduction

Male infertility has become a global problem because of its high incidence. According to statistics, infertile couples account for 15%–18% of couples of childbearing age, and male infertility accounts for 50% (Krausz and Riera-Escamilla, 2018). Spermatogenesis is a unique and complex process, and the blood-testis barrier (BTB) formed by the interaction among Sertoli cells and surrounding structures maintains the smooth functioning of the spermatogenesis process (Mruk and Cheng, 2015). Studies have found that substances that impair spermatogenic function can destroy the BTB, and proteins related to the BTB junction structure are the targets of various reproductive poisons. Glucosides of *Tripterygium wilfordii* Hook. f. (GTW) damages Sertoli cells and destroys the integrity of the BTB, which increases its permeability. These factors lead to a poor spermatogenic microenvironment and cause a decline in spermatogenic function (Xu et al., 2016).

In recent years, the role of traditional Chinese medicine (TCM) in the treatment of male infertility has been gradually confirmed (Jiang et al., 2017). The theory that “the kidney stores essence and governs the reproductive system,” found in “Huang Di’s Canon of Medicine” has always been a summary of the relationship between the kidney and reproductive function (Yu et al., 2019). The kidney promotes the growth and development and reproductive function, so the strength of kidney qi determines the fertility. The TCM syndrome of most men with oligoasthenospermia is kidney deficiency (Zhang et al., 2018), and the core treatment principle is tonifying the kidney and replenishing the essence. *Cuscuta japonica* Choisy is a unique parasitic vine that belongs to the family Convolvulaceae, and *Cuscuta* seed (Semen *Cuscutae*) has been widely used as a tonic to nourish the kidney and replenish the essence in TCM clinical practice. *Fructus Lycii* is the mature fruit of *Lycium barbarum* L. (Solanaceae). It has the same effect as *Semen Cuscutae*. *Semen Cuscutae*-*Fructus Lycii* (SC-FL) is the core herbal pair of many classic prescriptions for male infertility treatment, such as Zuogui Pill (Yang et al., 2018) and Wuzi Yanzong Pill (Zhao et al., 2019), which have the effect of tonifying the kidney and replenishing the essence. Until today, TCM physicians are used to combining the two herbs, and they have become frequently-used herbal pair to tonify the kidney and replenish the essence, which confirms the TCM theory that the kidneys govern the reproductive system (Guan et al., 2020).

Because most prescriptions contain multiple Chinese medicines, each Chinese medicine contains many ingredients, and traditional experimental methods are limited, the molecular targets and mechanism of most Chinese medicines are still unclear. Based on “disease-gene-target-drug” interactions, network pharmacology helps us to observe the interventions and influence of drugs on disease networks systematically and comprehensively by focusing on the concept of multilevel interaction networks with multiple points of view. We can scientifically explain the mechanism of Chinese medicine and the rules of the compatibility of prescriptions and discover new indications and new active compounds in medicines through network pharmacology.

Therefore, we used *Semen Cuscutae*-*Fructus Lycii* as the research object and observed its specific role and mechanism in repairing the BTB by in silico and in vitro methods to initially reveal the mechanism of the treatment of male infertility.

2. Methods

2.1. Study of network pharmacology

2.1.1. Data preparation and processing

2.1.1.1. Active ingredients and targets of *Semen Cuscutae*-*Fructus Lycii*. To determine the chemical ingredients of *Semen Cuscutae* (SC) and *Fructus Lycii* (FL), we performed a search of the Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform (TCMSP, <http://tcmsp.com/index.php>, updated on May 31, 2014) (Ru et al., 2014).

We used oral bioavailability (OB) $\geq 30\%$ and drug-like (DL) ≥ 0.18 for the properties of the compound as the screening conditions to identify the effective biologically active ingredients and corresponding targets of SC and FL. ADME is short for absorption, distribution, metabolism, and excretion. The early assessment of bioactive ingredients helps to choose ingredients with good pharmacokinetic properties, and OB and DL were as ADME-related models. The convergence of the ADME process is displayed by OB, which illustrated the percentage of an orally administered dose of the chemical ingredients that reaches the systemic circulation. The “drug-like” prospective of ingredients was estimated by DL. The protein name was converted into a gene name through the UniProt (<https://www.UniProt.org/>) database, and the selected species was set as “*Homo sapiens*”.

2.1.1.2. Targets related to male infertility. By taking “Male Infertility” as a keyword, we searched The Comparative Toxicogenomics Database (CTD, <http://ctdbase.org/>, update 2019) (Davis et al., 2019), Therapeutic Target Database (TTD, <http://db.idrblab.net/ttd/>, Update, 2020) (Wang et al., 2020), and the Online Mendelian Inheritance in Man (OMIM, <http://omim.org/>, updated on January 15, 2019) (Amberger et al., 2015), DisGeNET (<https://www.disgenet.org/>, updated in 2020) (Piñero et al., 2017) and GeneCards (<https://www.genecards.org/>, updated on January 31, 2019) databases (Safran et al., 2010). We determined the relevant targets for male infertility after summarizing the data and removing the duplicates.

2.1.1.3. Active ingredients and targets of GTW. By searching TCMSP, TTD, CTD, the DrugBank database (<https://www.drugbank.ca>, updated on December 20, 2018) (Wishart et al., 2018), and mining the literature, we determined the main chemical ingredients of GTW and identified its related targets. We queried the molecular structure of the compound against the PubChem chemical structure database and downloaded the structure diagram in SDF format. Next, we uploaded the SDF files to the PharmMapper server (<http://www.ilab-ecust.cn/pharmmapper/>, updated in 2017) (Wang et al., 2017), and the corresponding targets of the compound were predicted based on the pharmacophore. We merged and summarized the collected information about effective targets of GTW. The UniProt database was also used for name standardization.

2.1.2. Network construction

2.1.2.1. Screening of common targets for medicines and diseases and construction of the interaction network. R software (<https://www.r-project.org/>, Version 3.6.2) was used to determine the common targets of medicines and diseases. We determined the mutual intersections of SC-

Table 1

Active ingredient of Semen Cuscutae-Fructus Lycii retrieved from the TCMSP database.

No.	Molecule Name	OB	DL	No.	Molecule Name	OB	DL
1	Sitosterol alpha1	43.28	0.78	28	Lophenol	38.13	0.71
2	Cycloartenol	38.69	0.78	29	4alpha,14alpha,24-trimethylcholesta-8,24-dienol	38.91	0.76
3	Mandenol	42	0.19	30	4alpha,24-dimethylcholesta-7,24-dienol	42.65	0.75
4	Ethyl linolenate	46.1	0.2	31	4alpha-methyl-24-ethylcholesta-7,24-dienol	42.3	0.78
5	LAN	42.12	0.75	32	6-Fluorindole-7-Dehydrocholesterol	43.73	0.72
6	Stigmasterol	43.83	0.76	33	7-O-Methyluteolin-6-C-beta-glucoside_qt	40.77	0.3
7	beta-sitosterol	36.91	0.75	34	Atropine	42.16	0.19
8	atropine	45.97	0.19	35	Cryptoxanthin monoepoxide	46.95	0.56
9	campesterol	37.58	0.71	36	Cycloeucalenol	39.73	0.79
10	cyanin	47.42	0.76	37	(E,E)-1-ethyl octadeca-3,13-dienoate	42	0.19
11	24-methylidenelophenol	44.19	0.75	38	methyl (1R,4aS,7R,7aS)-4a,7-dihydroxy-7-methyl-1-[(2S,3R,4S,5S,6R)-3,4,5-trihydroxy-6-(hydroxymethyl)oxan-2-yl]oxy-1,5,6,7a-tetrahydrocyclopenta[d]pyran-4-carboxylate	39.43	0.47
12	daucosterol_qt	36.91	0.75	39	Lantadene A	38.68	0.57
13	glycitin	50.48	0.24	40	Physalin A	91.71	0.27
14	delta-Carotene	31.8	0.55	41	Phycion-8-O-beta-D-gentiobioside	43.9	0.62
15	CLR	37.87	0.68	42	lanost-8-en-3beta-ol	34.23	0.74
16	14 b-pregnane	34.78	0.34	43	lanost-8-enol	34.23	0.74
17	(24 R)-4alpha-Methyl-24-ethylcholesta-7,25-dien-3beta-ylacetate	46.36	0.84	44	Obtusifoliol	42.55	0.76
18	24-Methylenecycloartan-3beta,21-diol	37.32	0.8	45	quercetin	46.43	0.28
19	24-ethylcholest-22-enol	37.09	0.75	46	sesamin	56.55	0.83
20	24-ethylcholesta-5,22-dienol	43.83	0.76	47	NSC63551	39.25	0.76
21	24-methyl-31-norlanost-9(11)-enol	38	0.75	48	isorhamnetin	49.6	0.31
22	24-methylenelanost-8-enol	42.37	0.77	49	kaempferol	41.88	0.24
23	Fucosterol	43.78	0.76	50	campest-5-en-3beta-ol	37.58	0.71
24	31-Norcyclolaudenol	38.68	0.81	51	Isofucosterol	43.78	0.76
25	31-norlanost-9(11)-enol	38.35	0.72	52	matrine	63.77	0.25
26	31-norlanosterol	42.2	0.73	53	sophranol	55.42	0.28
27	4,24-methyllophenol	37.83	0.75	54	Cuscutoside B	47.56	0.18

FL targets, GTW targets and male infertility targets and used the Venn Diagram package to draw a Venn diagram. Then, we used Cytoscape (<http://www.cytoscape.org>, Version 3.8.0) software to build the medicine-active ingredient-target-disease interaction network. The nodes in the network graph represent diseases, medicines, active ingredients and targets. The connections of the edges represent the interactions and relationships between the active ingredient and the target and between the target and the disease.

2.1.2.2. Construction of the PPI network. The STRING (<https://string-db.org/>) (Szklarczyk et al., 2019) platform was used to build a common target protein-protein interaction (PPI) network. “*Homo sapiens*” was selected in the Organism column, and the minimum required interaction score was set to “medium confidence” (0.4). “Node” represents the different targets, and “edge” represents the relationship between different targets. Network Analysis, which is a plug-in for Cytoscape, was used to analyze the topological properties of the network, and the cytoHubba plug-in was used to screen the core targets.

2.1.2.3. Gene ontology (GO) enrichment analysis and Kyoto Encyclopedia of genes and genomes (KEGG) pathway enrichment analysis. The cluster-profiler package in R software was used to determine the GO and KEGG pathways enriched in the common targets. With a P value = 0.05 and a Q value = 0.05 as the thresholds, the results are presented in the form of a bar chart or bubble chart.

2.2. Experimental verification

2.2.1. Materials

2.2.1.1. Animals. Thirty-six specific pathogen-free (SPF grade) 12-week-old male Sprague-Dawley (SD) rats were purchased from SPF Laboratory Animal Technology Co., Ltd. (Laboratory Animal License Number: SCXK (Beijing) 2011-0004). The rats weighed 250 ± 20 g, and all had normal sexual function, which was confirmed with mating

experiments. The animal experiments were performed in an environmentally controlled room with a temperature of 23–25 °C and 50–80% humidity. Before the experiment started, the protocol was approved by the Animal Ethics Committee of Dongzhimen Hospital, Beijing University of Chinese Medicine (No: 2018–18).

2.2.1.2. Tested drug and reagents. The 10 mg GTW tablets (Batch No: 130801) were manufactured by Anhui Xinlonghai Pharmaceutical Co., Ltd (Suzhou, Anhui Province, China). Before administration, the GTW tablets were dissolved in deionized water and suspended at the desired concentration. The herb pair consisted of 15 g SC and 15 g FL, which were acquired from Dongzhimen Hospital, Beijing University of Chinese Medicine (Beijing, China), and purchased from Beijing KANGMEI Pharmaceutical Co., LTD (Beijing, China). Voucher specimen numbers for SC and FL were DC6171 and DCA011, respectively. Voucher specimens of SC and FL were deposited at the Key Laboratory of Dongzhimen Hospital, Beijing University of Chinese Medicine, Beijing, China. SC-FL (30 g) was decocted with 300 ml boiling deionized water for 60 min. The decoction was then filtered, collected and concentrated to one-sixth of the original volume using rotary evaporation at 60 °C. The dose conversion was in accordance with the standard dose for the animal experiment.

2.2.1.3. Main instruments. The following instruments were used: optical microscope (Olympus L340099); Weili sperm quality detection system (China, WLJY-9000); thermostatic water bath; precision electronic balance (Mettler AE160); pH Meter (Japan, TOA HM-20 E); cell counting board; scissors; tweezers; low temperature, high-speed centrifuge (Germany MIKRO2R); ACQUITY UPLC I-Class system (U.S.A, Waters).

2.2.2. Mass spectrometry analysis of SC-FL

15 g SC and 15 g FL were weighed, decocted with 100 ml boiling deionized water for 60 min we took a part of the decoction and added methanol at a ratio of 1:1. And then, the supernatant was obtained for ACQUITY UPLC analysis by centrifuging at 14,000 rpm/min for 20 min. Chromatography was performed on an ACQUITY UPLC BEH C18 column

Table 2

Targets of semen cuscutae-fructus lycii.

No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID
1	F7	2155	47	KCNH2	3757	93	VCAM1	7412	139	POR	5447	185	CXCL10	3627
2	PTGS2	5743	48	PRKACA	5566	94	NR1I2	8856	140	ODC1	4953	186	CHUK	1147
3	SCN5A	6331	49	CHRM3	1131	95	CYP1B1	1545	141	TOP1	7150	187	SPP1	6696
4	CCND1	595	50	CHRM1	1128	96	ALOX5	240	142	RAF1	5894	188	RUNX2	860
5	IL10	3586	51	GABRA2	2555	97	HAS2	3037	143	HIF1A	3091	189	RASSF1	11186
6	FASN	2194	52	CHRM4	1132	98	GSTP1	2950	144	RUNX1T1	862	190	E2F1	1869
7	ACACA	31	53	PDE3A	5139	99	AHR	196	145	HSPA5	3309	191	E2F2	1870
8	G6PD	2539	54	HTR2A	3356	100	PSMD3	5709	146	ERBB2	2064	192	ACPP	55
9	NOS3	4846	55	GABRA5	2558	101	SLC2A4	6517	147	CAV1	857	193	CTSD	1509
10	ECE1	1889	56	ADRA1D	146	102	NR1I3	9970	148	F3	2152	194	IGFBP3	3486
11	ACADM	34	57	GABRA3	2556	103	INSR	3643	149	GJA1	2697	195	IGF2	3481
12	CYP2B6	1555	58	CHRM2	1129	104	DIO1	1733	150	IL1B	3553	196	IRF1	3659
13	UGT1A8	54576	59	ADRA1B	147	105	PPP3CA	5530	151	CCL2	6347	197	ERBB3	2065
14	SREBF1	6720	60	ADRB2	154	106	GSTM1	2944	152	PTGER3	5733	198	PCOLCE	5118
15	NOX3	50508	61	CHRNA2	1135	107	GSTM2	2946	153	CXCL8	3576	199	NPEPPS	9520
16	NOX1	27035	62	SLC6A4	6532	108	AKR1C3	8644	154	PRKCB	5579	200	HK2	3099
17	ACOX1	51	63	OPRM1	4988	109	SLPI	6590	155	BIRC5	332	201	NKX3-1	4824
18	ACLY	47	64	BAX	581	110	NR3C2	4306	156	DUOX2	50506	202	RASA1	5921
19	EHHADH	1962	65	CASP9	842	111	ABAT	18	157	HSPB1	3315	203	NCOA1	8648
20	AUH	549	66	JUN	3725	112	ADH1B	125	158	SULT1E1	6783	204	ADRA2A	150
21	HADHB	3032	67	CASP3	836	113	ADH1C	126	159	MGAM	8972	205	SLC6A3	6531
22	DECR1	1666	68	CASP8	841	114	ADH1A	124	160	IL2	3558	206	LTA4H	4048
23	ECI1	1632	69	PRKCA	5578	115	LYZ	4069	161	CCNB1	891	207	MAOA	4128
24	PGR	5241	70	TGFB1	7040	116	MMP2	4313	162	PLAT	5327	208	CTRB1	1504
25	NOS2	4843	71	PON1	5444	117	IL6	3569	163	THBD	7056	209	ADRB1	153
26	PTGS1	5742	72	MAP2	4133	118	MYC	4609	164	SERPINE1	5054	210	CHRM5	1133
27	ESR1	2099	73	PRSS1	5644	119	HPSE	10855	165	COL1A1	1277	211	ADRA2C	152
28	AR	367	74	SERPIND1	3053	120	IER3IP1	51124	166	IFNG	3458	212	OPRD1	4985
29	PPARG	5468	75	CHRNA7	1139	121	CD44	960	167	PTEN	5728	213	HTR2C	3358
30	PTPN6	5777	76	SLC6A2	6530	122	AKR1B1	231	168	IL1A	3552	214	ADRA2B	151
31	ESR2	2100	77	TOP2B	7155	123	MMP3	4314	169	MPO	4353	215	DRD2	1813
32	DPP4	1803	78	PCP4	5121	124	RXRA	6256	170	TOP2A	7153	216	HTR1B	3351
33	MAPK14	1432	79	IKBKB	3551	125	EGFR	1956	171	ABCG2	9429	217	HRH1	3269
34	GSK3B	2932	80	AKT1	207	126	VEGFA	7422	172	NFE2L2	4780	218	HTR1A	3350
35	HSP90AA1	3320	81	CD40LG	959	127	BCL2L1	598	173	NQO1	1728	219	CHEK1	1111
36	CDK2	1017	82	AHSA1	10598	128	FOS	2353	174	PARP1	142	220	PIM1	5292
37	ACHE	43	83	MAPK8	5599	129	CDKN1A	1026	175	COL3A1	1281	221	CCNA2	890
38	GABRA1	2554	84	MMP1	4312	130	EIF6	3692	176	CXCL11	6373	222	APBA3	9546
39	MAOB	4129	85	STAT1	6772	131	PLAU	5328	177	CXCL2	2920	223	MMP13	4322
40	GRIA2	2891	86	CDK1	983	132	MMP9	4318	178	DCAF5	8816	224	MMP8	4317
41	RELA	5970	87	HMOX1	3162	133	MAPK1	5594	179	CHEK2	11200	225	NR3C1	2908
42	XDH	7498	88	CYP3A4	1576	134	EGF	1950	180	CLDN4	1364	226	DRD5	1816
43	NCF1	653361	89	CYP1A2	1544	135	RB1	5925	181	PPARA	5465			
44	OLR1	4973	90	CYP1A1	1543	136	TP53	7157	182	PPARD	5467			
45	NCOA2	10499	91	ICAM1	3383	137	ELK1	2002	183	HSF1	3297			
46	PIK3CG	5294	92	SELE	6401	138	NFKBIA	4792	184	CRP	1401			

Table 3
Targets related to male infertility.

No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID
1	AR	367	74	HSP90AA1	3320	147	PPARA	5465	220	FOXO3	2309	293	CDKN1C	1028
2	CYP1A1	1543	75	GCLC	2729	148	FASN	2194	221	FABP4	2167	294	ALDH6A1	4329
3	ESR1	2099	76	NR1I2	8856	149	NR3C1	2908	222	ALDH1A1	216	295	CREB1	1385
4	BAX	581	77	PTGS2	5743	150	NTRK2	4915	223	LPL	4023	296	MAPT	4137
5	BCL2	596	78	IGFBP2	3485	151	CP	1356	224	EEF2	1938	297	ACADM	34
6	LHB	3972	79	SGK1	6446	152	IGF2	3481	225	HSD11B1	3290	298	CASP7	840
7	TP53	7157	80	MMP2	4313	153	NFKB1	4790	226	CCNB1	891	299	SP1	6667
8	FSHB	2488	81	CTSD	1509	154	HSPD1	3329	227	PRL	5617	300	SQSTM1	8878
9	AHR	196	82	GSTA4	2941	155	ABCC3	8714	228	DUSP6	1848	301	ARG1	383
10	SOD2	6648	83	MKI67	4288	156	APOE	348	229	GAS6	2621	302	SQLE	6713
11	CYP17A1	1586	84	PARP1	142	157	BCHE	590	230	INHBB	3625	303	DIO2	1734
12	SELENOP	6414	85	EGFR	1956	158	CFL1	1072	231	SLC2A4	6517	304	ANPEP	290
13	MDM2	4193	86	DNMT3A	1788	159	SRD5A1	6715	232	PDIA3	2923	305	GCLM	2730
14	GPX4	2879	87	CYP2E1	1571	160	LEP	3952	233	CXCL10	3627	306	TXNIP	10628
15	ALDH2	217	88	CDH1	999	161	ATF4	468	234	G6PD	2539	307	FSHR	2492
16	ENO1	2023	89	SERPINE1	5054	162	NGF	4803	235	AKT2	208	308	NFIX	4784
17	NOS3	4846	90	CTSB	1508	163	CTSH	1512	236	MGLL	11343	309	CAPN2	824
18	TSC22D3	1831	91	DNMT1	1786	164	MCM2	4171	237	PTTG1	9232	310	GLRX	2745
19	LHCGR	3973	92	GADD45A	1647	165	FAS	355	238	ALAD	210	311	TGFB1	7045
20	GJA1	2697	93	CTNNB1	1499	166	TGFB1	7040	239	CCNA2	890	312	CFB	629
21	STAR	6770	94	ABCC2	1244	167	STAT3	6774	240	TRIB3	57761	313	PSMB5	5693
22	CAT	847	95	HSP90AB1	3326	168	VEGFA	7422	241	CTSC	1075	314	AURKA	6790
23	CASP3	836	96	GADD45G	10912	169	HSD17B3	3293	242	ME1	4199	315	PCK2	5106
24	CCND1	595	97	ATF3	467	170	PPARGC1A	10891	243	PRKCA	5578	316	TGFB3	7043
25	GSR	2936	98	GLUL	2752	171	GADD45B	4616	244	LDHA	3939	317	MAPK11	5600
26	FOS	2353	99	CALR	811	172	NR4A1	3164	245	SCARB1	949	318	S100A6	6277
27	TNF	7124	100	CEBPA	1050	173	NFKBIA	4792	246	TTR	7276	319	PMP22	5376
28	EPHX1	2052	101	C3	718	174	NOS2	4843	247	S100A9	6280	320	TPM3	7170
29	CDK1	983	102	GPX3	2878	175	HMGCR	3156	248	GHR	2690	321	CXCR4	7852
30	HMOX1	3162	103	CD36	948	176	IER3	8870	249	INHBA	3624	322	TNFRSF11B	4982
31	IGF1	3479	104	SLC40A1	30061	177	SLC3A2	6520	250	TNFSF10	8743	323	MTHFD2	10797
32	PPARG	5468	105	CDC20	991	178	LIF	3976	251	KRT19	3880	324	CDKN3	1033
33	NFE2L2	4780	106	DNMT3B	1789	179	CEBPB	1051	252	CD44	960	325	ACACA	31
34	GSTM1	2944	107	CAV1	857	180	ALDOA	226	253	ID3	3399	326	HRAS	3265
35	MAPK1	5594	108	DDIT3	1649	181	PRDX1	5052	254	KRT8	3856	327	ABCG1	9619
36	VIM	7431	109	GSTT1	2952	182	B2M	567	255	CXCL8	3576	328	CCL5	6352
37	GPX1	2876	110	GSK3B	2932	183	ABCC4	10257	256	GOT1	2805	329	CD59	966
38	IL4	3565	111	PGR	5241	184	HPRT1	3251	257	IGFBP4	3487	330	FTH1	2495
39	HSP90B1	7184	112	MGP	4256	185	CCNE1	898	258	PRLR	5618	331	CFTR	1080
40	IL6	3569	113	PRDX6	9588	186	GSTA3	2940	259	TSHB	7252	332	AURKC	6795
41	CDKN1A	1026	114	S100A8	6279	187	ACAA2	10449	260	EGF	1950	333	NR5A1	2516
42	ACTB	60	115	FN1	2335	188	ABCG2	9429	261	CASP1	834	334	CATSPER1	117144
43	GSTP1	2950	116	IGFBP3	3486	189	JUNB	3726	262	HMGCS1	3157	335	GNRH1	2796
44	CYP11A1	1583	117	MT2A	4502	190	ACTA2	59	263	TGFB2	7042	336	SRY	6736
45	SOD1	6647	118	FABP3	2170	191	BIRC5	332	264	GPC1	2817	337	AMH	268
46	CLU	1191	119	TPM1	7168	192	RELA	5970	265	MAPK8	5599	338	DAZ1	1617
47	MAPK3	5595	120	AKT1	207	193	TF	7018	266	HES1	3280	339	DNAH1	25981
48	IFNG	3458	121	GSN	2934	194	LCN2	3934	267	RB1	5925	340	NR0B1	190
49	CASP9	842	122	CCL2	6347	195	ABCB1	5243	268	DBP	1628	341	SOX9	6662
50	HSPB1	3315	123	SPP1	6696	196	NR1I3	9970	269	CYCS	54205	342	DDX3Y	8653
51	EGR1	1958	124	CYP3A4	1576	197	KLF9	687	270	IGFBP5	3488	343	STRC	161497
52	NQO1	1728	125	OCLN	100506658	198	CCN2	1490	271	ANXA1	301	344	WDR66	144406
53	CYP1A2	1544	126	BHLHE40	8553	199	GSTM2	2946	272	TOP2A	7153	345	USP9Y	8287
54	IL1B	3553	127	CXCL12	6387	200	ID1	3397	273	HADH	3033	346	DAZ4	57135

(continued on next page)

Table 3 (continued)

No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID
55	PCNA	5111	128	IL10	3586	201	CEBPD	1052	274	FST	10468	347	WT1	7490
56	HSPA5	3309	129	CCND2	894	202	LGALS1	3956	275	PK3R1	5295	348	DAZ2	57055
57	CYP19A1	1588	130	GSTM3	2947	203	NAMPT	10135	276	ATP2A2	488	349	CATSPER2	117155
58	JUN	3725	131	ALAS1	211	204	PTEN	5728	277	IL1R1	3554	350	SPATA16	83893
59	ATP1B1	481	132	CCND3	896	205	NDRG1	10397	278	CCNB2	9133	351	DAZ3	57054
60	ESR2	2100	133	LGALS3	3958	206	COL1A1	1277	279	TIMP2	7077	352	SRD5A2	6716
61	MYC	4609	134	MGST1	4257	207	ANXA3	306	280	DUSP1	841	353	INS	3630
62	APP	351	135	GPT	2875	208	ALPL	249	281	CASP8	841	354	CYP21A2	1589
63	GSTA1	2938	136	TJP1	7082	209	GPX2	2877	282	EDN1	1906	355	INSL3	3640
64	HSPA8	3312	137	IGF1R	3480	210	IL2	3558	283	FABP5	2171	356	BCL10	8915
65	GSTA2	2939	138	PDGFRA	5156	211	ATP1A1	476	284	GPRI1	2852	357	MAK	4117
66	BCL2L1	598	139	IRS1	3667	212	POR	5447	285	ACAT1	38	358	ME1	4201
67	GAPDH	2597	140	BTG2	7832	213	TK1	7083	286	MCM6	4175	359	BRCA2	675
68	HSPA1A	3303	141	ACHE	43	214	TIMP3	7078	287	STAT1	6772	360	ACR	49
69	CYP1B1	1545	142	ID2	3398	215	PDI4	9601	288	PRDX2	7001	361	ALG13	79868
70	THRA	7067	143	HSPA9	3313	216	GPXMB	10457	289	SELENP1	8991	359	BRCA2	675
71	CDH2	1000	144	TKT	7086	217	MCMS	4174	290	DDIT4	54541	360	ACR	49
72	CLDN1	9076	145	TFR3	7037	218	PKM	5315	291	FKBP5	2289	361	ALG13	79868
73	BDNF	627	146	ALB	213	219	CPT1A	1374	292	TFF1	7031			

(150 × 2.1 mm, 1.7 μm, Waters Corporation, Milford, MA, U.S.A) at a flow rate of 0.3 mL/min and a 40 °C column temperature. The volume of each injection was 3 μL. The mobile phase consisted of 0.1% acetonitrile formate (A) and 0.1% water formate (B): 0–12 min, 99%–75% B; 12–15min, 75%–50% B; 15–18 min, 50%–10% B; 18–20min, 10%–1% B; 20–22min, 1% B. TurboIon Spray ion source and ESI positive and negative ion scanning mode were employed to perform the Time of Flight Mass Spectrometry. The conditions were as follows: Ion Source Gas1 (Gas1):55, Ion Source Gas2 (Gas2): 45, Curtain gas (CUR):35, source temperature: 600 °C, IonSapary Voltage Floating (ISVF): 5500 V/-4500 V (positive and negative ion modes); TOF MS scan m/z range:50–1500 Da, production scan m/z range: 25–1500 Da, TOF MS scan accumulation time 0.25 s/spectra, product ion scan accumulation time 0.035 s/spectra; the secondary MS was acquired by Information-Dependent Acquisition (IDA) with high sensitivity mode, Declustering potential (DP): ±60 V (positive and negative ion modes); Collision Energy: 35 ± 15 eV. IDA set were as follows: Exclude isotopes within 4 Da, Candidate ions to monitor per cycle: 12. The target compounds were identified by SCIEX OS software 1.4 based on the first-order accurate mass number, isotope distribution ratio, and MS/MS of the compounds.

2.2.3. Spermatogenesis dysfunction rat model

Thirty-six SD rats were randomly divided into the NC group, SD group and SCFL group (n = 12). In the first four weeks of the experiment, rats in the SD and SCFL groups were intragastrically administered the GTW suspension (10 ml/kg/day) (1 ml contained 4 mg GTW, which is 40 times the amount used in humans). Rats in the NC group were treated with deionized water (10 ml/kg) for 28 days.

2.2.4. Treatment

During weeks 5–8 of the experiment, the rats of the NC and SD groups were intragastrically administered deionized water (10 ml/kg/day), and those of the SCFL group were intragastrically administered the SC-FL suspension (10 ml/kg/day) daily. Each milliliter of the decoction contained 0.6 g of SC-FL (equivalent to 3 g/kg of herbal medicine).

2.2.5. Preparation of the tissue samples

After four weeks of treatment, the rats were anesthetized with 2% pentobarbital sodium (50 mg/kg, i.p.), and the testis and epididymal tissues were harvested and weighed with an electronic balance. The epididymis was used to detect sperm quality. The left testis was used to detect protein and mRNA expression by Western blotting and PCR, respectively, and the right testis was used for morphological observations.

2.2.6. Determination of sperm quality

Each epididymis was transferred into an Eppendorf tube containing M2 working solution at 37 °C, and we operated on a table with a constant temperature of 37 °C. The epididymis was transferred into 200 μl of balanced G-IVF Plus, and the creamy sperm were squeezed out using tweezers. When the sperm were fully dispersed, we transferred them into centrifuge tubes together with the epididymal tubes. After centrifugation at 300 rpm for 5 min, the tubes were placed in an incubator at 37 °C with 6% CO₂ for 30 min. We removed the middle layer of the liquid and added it to the sperm counting board, and the sperm count and motility were determined according to the semen analysis method described in the WHO laboratory manual for the examination and processing of human semen.

2.2.7. Determination of the morphology of testicular tissue

Tissue paraffin embedding: Testicular tissues were immersed in 4% paraformaldehyde for 24 h and then dehydrated with an alcohol gradient. After xylene transparency treatment, each sample was routinely embedded in paraffin and sliced. Hematoxylin-eosin (H&E) staining: Slides were deparaffinized with a gradient of xylene and alcohol, stained with hematoxylin for 3 min, and differentiated in

Table 4

Targets of GTW.

No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID
1	NR3C1	2908	46	CXCL8	3576	91	BAK1	578	136	RHOB	388
2	VEGFA	7422	47	MCL1	4170	92	CLDN11	5010	137	RIPK1	8737
3	CCND1	595	48	IL2	3558	93	G6PC	2538	138	ROCK1	6093
4	BCL2	596	49	IFNG	3458	94	GJA1	2697	139	SLC2A2	6514
5	BCL2L1	598	50	IL4	3565	95	IL6	3569	140	STAR	6770
6	BAX	581	51	CD40	958	96	MLKL	197259	141	TAT	6898
7	FLT1	2321	52	CD14	929	97	NOTCH3	4854	142	TNNI3	7137
8	KDR	3791	53	C3	718	98	OCLN	100506658	143	BMP2	650
9	MMP9	4318	54	VTCN1	79679	99	PCK1	5105	144	CASP7	840
10	JUN	3725	55	IL1B	3553	100	TJP1	7082	145	ALB	213
11	PTGS2	5743	56	TNF	7124	101	TXN	7295	146	APOA2	336
12	BIRC5	332	57	TNFSF10	8743	102	CDK2	1017	147	STS	412
13	BIRC2	329	58	HMOX1	3162	103	CXCL1	2919	148	MAPKAPK2	9261
14	CFLAR	8837	59	AKT1	207	104	CYP11A1	1583	149	AKR1C2	1646
15	AKR1B1	231	60	AR	367	105	GCLC	2729	150	CA2	760
16	CDC37	11140	61	CASP9	842	106	NGF	4803	151	PPIA	5478
17	MMP2	4313	62	HSPA1A	3303	107	NR1H4	9971	152	GSTP1	2950
18	TIMP1	7076	63	ICAM1	3383	108	PPARA	5465	153	PGR	5241
19	TIMP2	7077	64	PTEN	5728	109	PPARGC1A	10891	154	BACE1	23621
20	MMP1	4312	65	VCAM1	7412	110	PTK2	5747	155	CA1	759
21	MYC	4609	66	EGFR	1956	111	RHOA	387	156	KIF11	3832
22	FN1	2335	67	MPO	4353	112	SIRT1	23411	157	MMP13	4322
23	COL7A1	1294	68	PARP1	142	113	SLC10A1	6554	158	MAOB	4129
24	TGFB1	7040	69	SELE	6401	114	TXNIP	10628	159	GC	2638
25	COL4A4	1286	70	TNFRSF10A	8797	115	ACTB	60	160	MTOR	2475
26	XIAP	331	71	TNFRSF10B	8795	116	BUB1	699	161	APIP	51074
27	DEFB4A	1673	72	ADGRE1	2015	117	CCL11	6356	162	RUFY3	22902
28	CD80	941	73	CASP8	841	118	CCNB1	891	163	MDM2	4193
29	CD86	942	74	CEBPA	1050	119	CDC42	998	164	EEF1A1	1915
30	CXCR4	7852	75	DHODH	1723	120	CDK1	983	165	MEX3D	399664
31	BIRC3	330	76	FBXO32	114907	121	CRK	1398	166	EIF2AK2	5610
32	CD274	29126	77	GPT	2875	122	CXCL2	2920	167	RAC1	5879
33	IL23A	51561	78	IL10	3586	123	DNM1L	10059	168	ANP32B	10541
34	CCR7	1236	79	IL17A	3605	124	DUSP1	1843	169	EGF	1950
35	CD1A	909	80	MAPK1	5594	125	GPX1	2876	170	ACBD7	414149
36	RELA	5970	81	MAPK3	5595	126	GPX2	2877	171	F11	2160
37	STAT3	6774	82	NFE2L2	4780	127	GPX3	2878	172	HK1	3098
38	FOS	2353	83	PPARG	5468	128	GPX4	2879	173	PFN1	5216
39	CDKN1A	1026	84	RPS6KB1	6198	129	GUSB	2990	174	RAB5C	5878
40	PLAU	5328	85	SLC7A11	23657	130	IL12B	3593	175	SRP19	6728
41	CD40LG	959	86	TLR6	10333	131	MFN2	9927	176	LHX4-AS1	100527964
42	CASP3	836	87	NOTCH1	4851	132	NPHS1	4868	177	ACOX1	51
43	TP53	7157	88	NQO1	1728	133	NPHS2	7827	178	CYB5B	80777
44	MAPK8	5599	89	HES1	3280	134	NROB2	8431	179	ACACB	32
45	STAT1	6772	90	APAF1	317	135	POR	5447	180	TIMM9	26520

Table 5

Common targets of drugs and diseases.

No.	SYMBOL	ENTREZID	No.	SYMBOL	ENTREZID
1	PTGS2	5743	23	EGFR	1956
2	CCND1	595	24	VEGFA	7422
3	IL10	3586	25	BCL2L1	598
4	PGR	5241	26	FOS	2353
5	AR	367	27	CDKN1A	1026
6	PPARG	5468	28	MAPK1	5594
7	RELA	5970	29	EGF	1950
8	BAX	581	30	TP53	7157
9	CASP9	842	31	POR	5447
10	JUN	3725	32	GJA1	2697
11	CASP3	836	33	IL1B	3553
12	CASP8	841	34	CXCL8	3576
13	TGFB1	7040	35	BIRC5	332
14	AKT1	207	36	IL2	3558
15	MAPK8	5599	37	CCNB1	891
16	STAT1	6772	38	IFNG	3458
17	CDK1	983	39	PTEN	5728
18	HMOX1	3162	40	NFE2L2	4780
19	GSTP1	2950	41	NQO1	1728
20	MMP2	4313	42	PARP1	142
21	IL6	3569	43	PPARA	5465
22	MYC	4609	44	NR3C1	2908

hydrochloric acid in alcohol for 2 s. The slides were washed under running water for 20 min and then stained with eosin for 2 min, followed by incubation in 95% alcohol for 3 min, 100% alcohol for 3 min, and xylene for 10 min. The slides were sealed with resin and dried in a fume hood.

2.2.8. Determination of the ultrastructure of testicular tissue

Testicular tissue was cut to a size of 1 mm³, fixed with 3% glutaraldehyde and stored at 4 °C. Then, the tissue was removed and rinsed 3 times with PBS. After fixation with 1% osmic acid at 4 °C for 2 h, the testicular tissue was washed with water for 5 min and rinsed with double-distilled water at 4 °C three times for 10 min each. The next step was acetone gradient dehydration. The tissue was soaked in acetone percolate (1, 2, 3) and placed at the bottom of the embedding capsule. We added embedding medium to the top of the capsule and dried it in the oven. We sliced the tissue using a Leica freezing microtome. The tissue was dyed with uranium acetate and citric acid, washed, dried and sliced. Slides were mounted and visualized under a microscope to find the seminiferous tubules first. Then, we adjusted the magnification to observe the Sertoli cells, spermatogenic cells, and BTB junction structure.

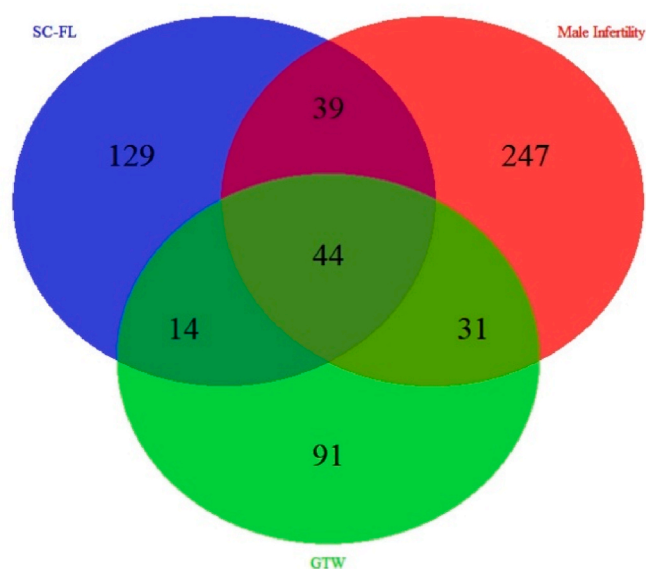


Fig. 1. Venn diagram of drug and disease targets. The blue represents the targets of Semen Cuscutae-Fructus Lycii, the red represents the targets related to male infertility, and the green represents targets of GTW. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

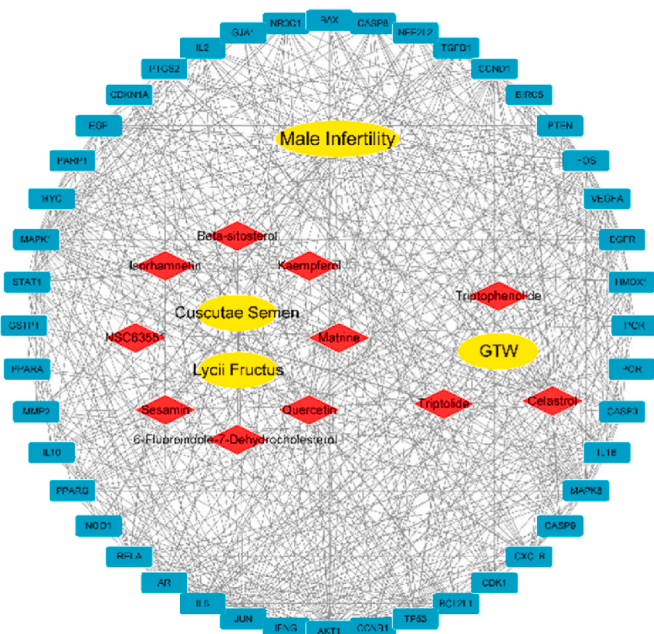


Fig. 2. Network of medicine-active ingredient-target-disease. Yellow oval, red diamond, and blue rectangle represent diseases/medicine, active ingredient, and targets, respectively, and black lines indicate mutual relationships. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.2.9. Protein expression of *c-jun*, *p38 MAPK*, *Occludin*, *ZO-1*, β -catenin, *N-cadherin* and *connexin 43* in the testis

①Total protein extraction: The solution consisted of 100 μ l RIPA buffer and 1 μ l protease inhibitor was used to fully grind and lyse 2 g testicular tissue. The homogenized solution was placed on ice for 30

min, and then centrifuged at 4 °C for 12,000 rpm for another 10 min. The supernatants were collected and 20 μ l Loading Buffer was added. The solution was boiled for 10min in order to denature the protein, and cooled to room temperature. We stored the final solution at –80 °C for later use.

②Western blotting detection: The protein sample subjected to polyacrylamide gel electrophoresis (SDS-PAGE), with the sample volume of 10 μ l per well. After the SDS-PAGE electrified in the condition of 80 V 40 min and 120 V 20 min, the sample was transferred to a polyvinylidene difluoride (PDVF) membrane, and then 280 mA for 1 h. The primary antibody was added after blocking with milk for 2 h (the antibody was diluted with milk, and the concentration was 1:1000–1:10000 in accordance with the instructions), and the samples were incubated at 4 °C overnight. Next, the membrane was incubated for 1 h with the secondary antibody (1XTBST, 1:2000) at room temperature after washing with 1XTBST three times. Then, the membrane was washed with 1XTBST three times again, and ECL developer was added. Quantitative analysis of the protein bands was performed using QuantityOne software. The relative amount of each protein was expressed as the ratio of the density of the target protein band to that of the β -tubulin band.

2.2.10. mRNA expression of *c-jun*, *p38 MAPK*, *occludin*, *ZO-1*, β -catenin, *N-cadherin* and *connexin 43* in the testis

①Total RNA extraction: We weighed the testicular tissue and quickly mashed it evenly. All steps were completed at a temperature of about 0 °C. Total RNA was extracted from the tissue using the TRIzol reagent according to the manufacturer's instructions. The mixture was centrifuged and homogenized, and placed at room temperature for 15 min. The supernatant was obtained and 0.2 ml chloroform was added. We shook the centrifuge tube vigorously for 10min, and then 4 °C, 12000 rpm, 10min. We removed the supernatant, and recorded the volume. Then we added an equal volume of isopropanol, and placed the tube at room temperature for 10min, and then 4 °C, 12000 rpm, 10min. We removed the supernatant, and added 75% ethanol, 4 °C, 12000 rpm, 5min, repeat this step twice. After removing the supernatant, RNA dissolved in DEPC-treated ultrapure water. At last, The RNA purity and concentration were measured using a Nanodrop2000.

②Reverse transcription reaction: FastKing RT Kit (With gDNase) kit (TIANGEN, China) was used for reverse transcription reaction. a. gDNA removal reaction system: 5 $\times$ gDNA Buffer 2 μ l, Total RNA 1–2 μ l, RNase-Free ddH₂O to make up to 10 μ l, mixed thoroughly, centrifuged and placed at 42 °C for 3 min. Then we placed the tube on ice; b. Reverse transcription reaction system: 10 $\times$ King RT Buffer 2 μ l, FastKing RT Enzyme Mix 1 μ l, FQ-RT Primer Mix 2 μ l, RNase-Free ddH₂O to make up to 10 μ l. We mixed the solution of a. and b. system thoroughly, incubated at 42 °C for 15 min, 95 °C for 3 min and placed on ice. The obtained cDNA could be used for subsequent experiments or stored at low temperature.

③qRT-PCR reaction: POWRUP SYBR MASTER MIX Kit (ABI, USA) was used for qRT-PCR reaction. a. Reaction system 20 μ l: 2x miRcute Plus miRNA PreMix 10 μ l, Forward Primer 0.4 μ l, Reverse Primer 0.4 μ l, cDNA 1 μ l, RNase-Free ddH₂O supplemented to 20 μ l; b. qPCR reaction program: 95 °C 15min; 95 °C 10sec; 60 °C 30sec, 40 cycles; Melting/Dissociation Curve Stage. At the end of the reaction, we recorded the number of cycles (Ct value) required for the fluorescence signal to reach the set threshold in each tube. The relative quantification (RQ) method ($RQ = 2^{-\Delta\Delta Ct}$) was used to calculate the expression of the target gene relative to that of the reference gene to compare the mRNA expression differences.

2.2.11. Statistical analyses

SPSS v21.0 statistical software (SPSS Inc., Chicago, Illinois, USA) was used for to analyze and process the data. Data were expressed as the

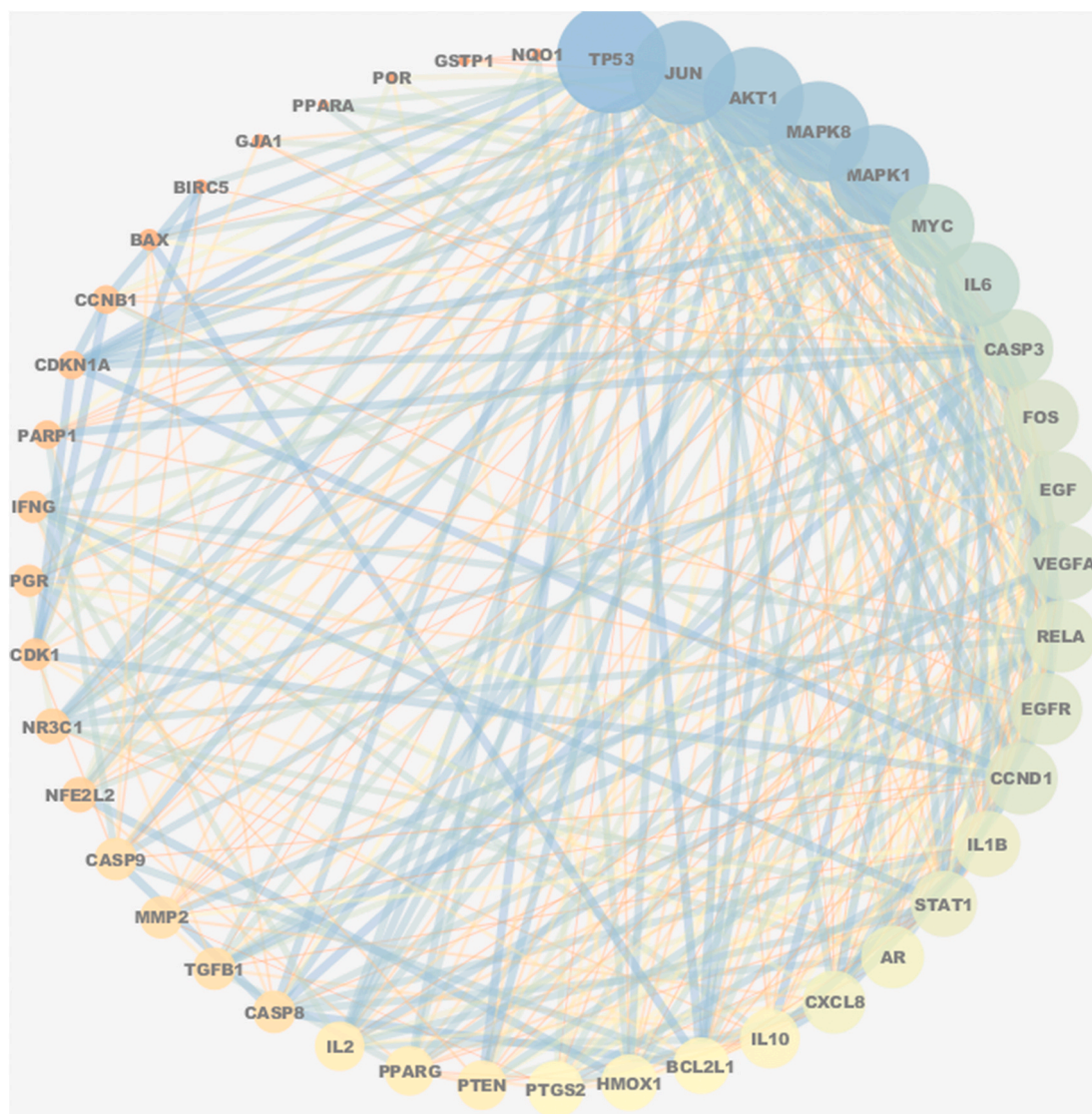


Fig. 3. PPI network. The oval represents the target point, and the straight line represents the relationship. The size and color of nodes in the network base on the Degree value, and the thickness and color of the edges base on the combined_score value (Low values to small sizes, Low values to bright colors). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

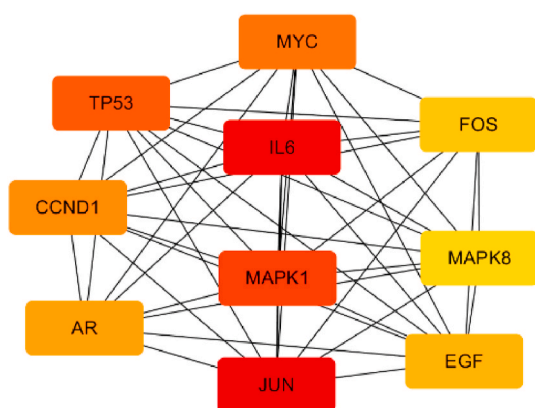


Fig. 4. Hub Gene. Rectangles represent genes, and straight lines represent relationships.

means $\pm$ SD. First of all, the normality test and the homogeneity of variance test were performed. If the data were normally distributed, the one-way ANOVA was used for multiple comparisons; If the data conformed to the homogeneity of the variance, the LSD test and SNK test were performed for multiple comparisons; If the data variance was not homogeneous, Welch analysis was used for pairwise comparison, and Games-Howell test was used for multiple comparisons; If the data didn't conform to the normal distribution, a nonparametric test was used. $P < 0.05$ was considered statistically significant.

3. Results

3.1. Active ingredients and targets of *Semen Cuscutae-Fructus Lycii*

A total of 213 chemical ingredients of SC and FL were retrieved from the TCMSP database. The screening conditions were $OB \geq 30\%$ and $DL \geq 0.18$, and 54 effective chemical ingredients were obtained, as shown in Table 1. After merging and removing the duplicates, 226 corresponding targets were obtained, as shown in Table 2.

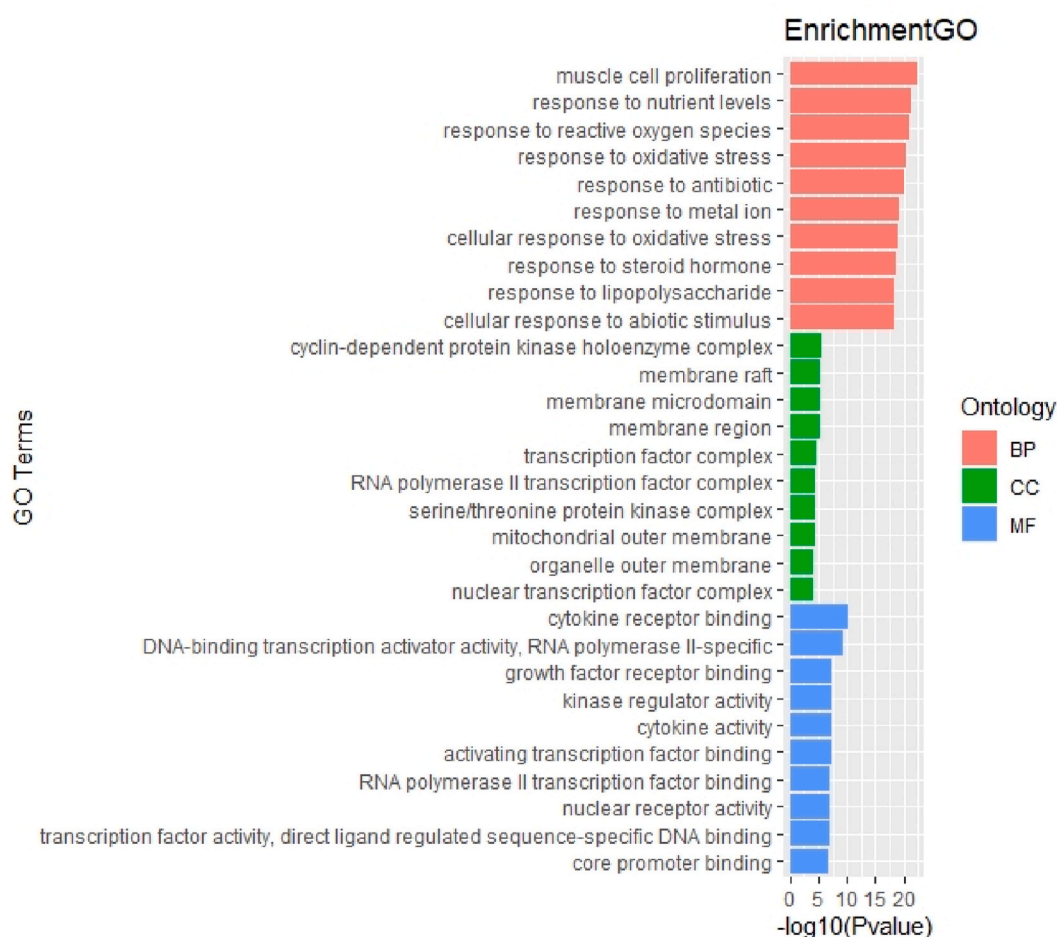


Fig. 5. GO enrichment analysis of common targets.

3.2. Targets related to male infertility

We searched CTD, TTD, OMIM, DisGeNET and GeneCards. We screened the top 1% of targets (≥ 0.42) in DisGeNET according to Score_gda, the top 1% of targets (≥ 105.52) in CTD according to the Inference Score, and the top 1% of targets (≥ 37.66) in GeneCards according to the Relevance score. After merging and removing duplicates, 361 target genes were obtained, as shown in Table 3.

3.3. Active ingredients and targets of GTW

According to a database search and literature mining, seven effective chemical ingredients, triptolide, tryptophanamide, celastrol, wilforine, wilfordine and wilfortrine were identified (Du et al., 2019; Du et al., 2018). We further searched the database and predicted the corresponding targets by means of chemical structure pharmacophores. The CTD database was screened according to an Interaction Count ≥ 2 , and Pharmmapper was screened according to a Normalized Fit Score ≥ 0.9 . After merging and removing the duplicates, 180 targets were obtained, as shown in Table 4.

3.4. Screening of common targets of medicines and diseases and construction of the interaction network

The intersections of the SC-FL targets, male infertility targets and GTW targets were determined to obtain 44 common targets, and a Venn diagram was drawn, as shown in Table 5 and Fig. 1. We imported the file into Cytoscape to draw the SC-FL-active ingredients-targets-GTW-male

infertility interaction network diagram, as shown in Fig. 2.

3.5. Construction of the PPI network

A PPI network of the targets of SC-FL for treating male infertility was constructed by using STRING. There were 45 nodes in the interaction network, and the number of edges was 364. The size and color of the nodes in the network were based on Degree Value, and the thickness and color of the edges were determined according to the combined score value (low values correspond to small sizes and bright colors). The higher the degree value is, the more critical the role of the target is. The higher the score is, the stronger the interaction between the two targets is, as shown in Fig. 3. We imported the PPI network into Cytoscape and used the latest MCC algorithm in the cytoHubba plugin to obtain the top 10 core targets, as shown in Fig. 4.

3.6. GO enrichment analysis and KEGG pathway enrichment analysis

GO enrichment analysis and KEGG pathway enrichment analysis were used to explore the biological functions of the common targets of SC-FL in male infertility. GO enrichment analysis includes three types: biological process (BP), cellular component (CC), and molecular function (MF); after GO enrichment, 21, 49, and 1487 results were obtained, respectively, and the top 10 results for each enrichment type were selected. See Fig. 5. The main BPs were muscle cell proliferation, response to reactive oxygen species, response to oxidative stress, and cellular response to oxidative stress. The main CCs were the cyclin-dependent protein kinase holoenzyme complex, membrane raft,

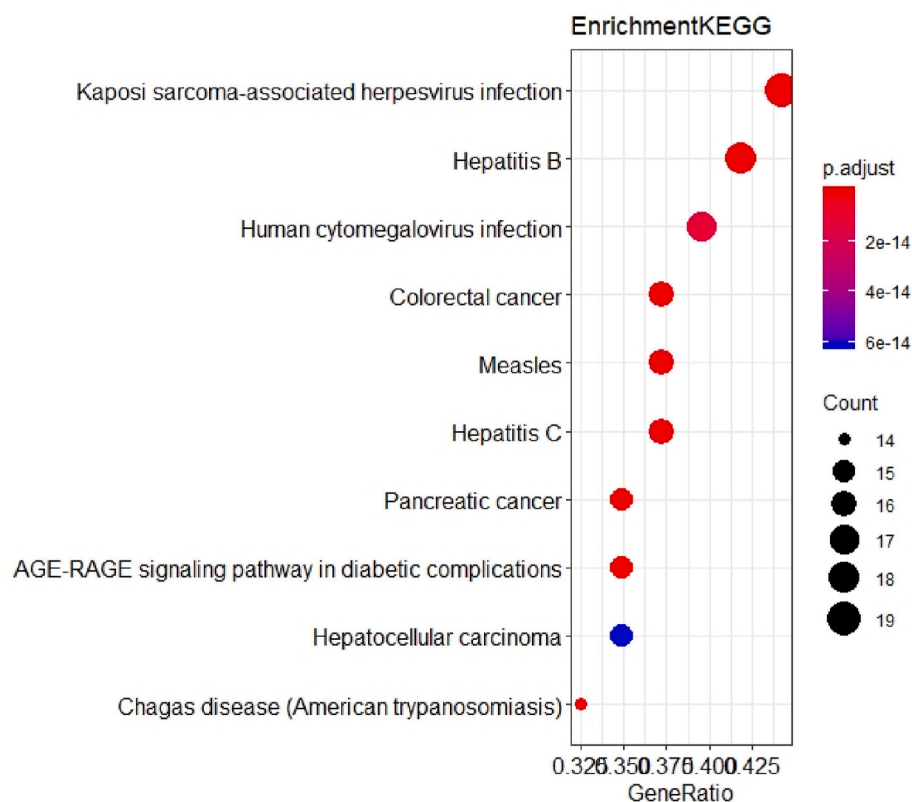


Fig. 6. The Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis of common targets. Red ellipses represent the selected hub genes in each module. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

membrane microdomain, and RNA polymerase II transcription factor complex. The main MFs are cytokine receptor binding, DNA-binding transcription activator activity, RNA polymerase II-specific, growth factor receptor binding, skin regulator activity, and skin activator activity. After KEGG enrichment analysis, 122 related signaling pathways were identified. The results are shown in Fig. 6, and the top 22 pathways were selected, as shown in Table 6. The main related pathways include the PI3K-Akt signaling pathway, apoptosis pathway, MAPK signaling pathway, and cellular senescence pathway.

3.7. Chemical ingredients and UHPLC-Q/TOF-MS analysis of water decoction of SC-FL

The total ion flow diagram of SC-FL was acquired from UHPLC-Q/TOF-MS analysis, and then the SCIEX OS software was used to collect and process data. Multiple confidence criteria including mass accuracy, retention time, isotope, and most importantly, library matching were used (He et al., 2016). Secondary TCM MS/MS Library containing secondary data of more than 1700 ingredients of TCM configured by SCIEX OS were searched based on the first-level accurate mass, isotope distribution ratio and MS/MS of the ingredient. This method aimed to complete the screening of the target without standard products. Part of ingredients were shown in Fig. 7 and Table 7.

3.8. The sperm concentrations and PR (progressive motility) +NP (nonprogressive motility)

The sperm concentrations of the SD group were significantly lower than those of the NC group ($P < 0.01$), while those of the SCFL group were significantly higher than those of the SD group ($P < 0.01$). The percentages of PR (progressive motility) +NP (nonprogressive motility) of the SD group were significantly lower than those of the NC group ($P <$

0.01), while those of the SCFL group were significantly higher than those of the SD group ($P < 0.01$). The results are shown in Figs. 8 and 9.

3.9. Testicular tissue morphology

It can be seen under the microscope that the structure of the seminiferous tubules of rats in the NC group was normal and orderly arranged, and the lumen structure was intact. There were a large number of Sertoli cells and sperm cells in the lumens, the structure of cells was complete, and the gaps between cells were tight. A large number of seminiferous tubules with abnormal structures were observed in the SD group. The lumens were shrunken and thin and intricately arranged, and the gap between the tubules was increased. There were many cavity-type seminiferous tubules, and there were a few Sertoli cells and sperm cells in the lumen with an abnormal shape and structure. The morphology of rat testicular tissue in the SCFL group was healthier than that of rats in the SD group, as the lumens were larger, and the number of Sertoli cells and sperm counts were significantly higher. Moreover, some seminiferous tubules had recovered and appeared more highly ordered, but when compared with those of rats in the NC group, the structures had not fully recovered, as shown in Fig. 10.

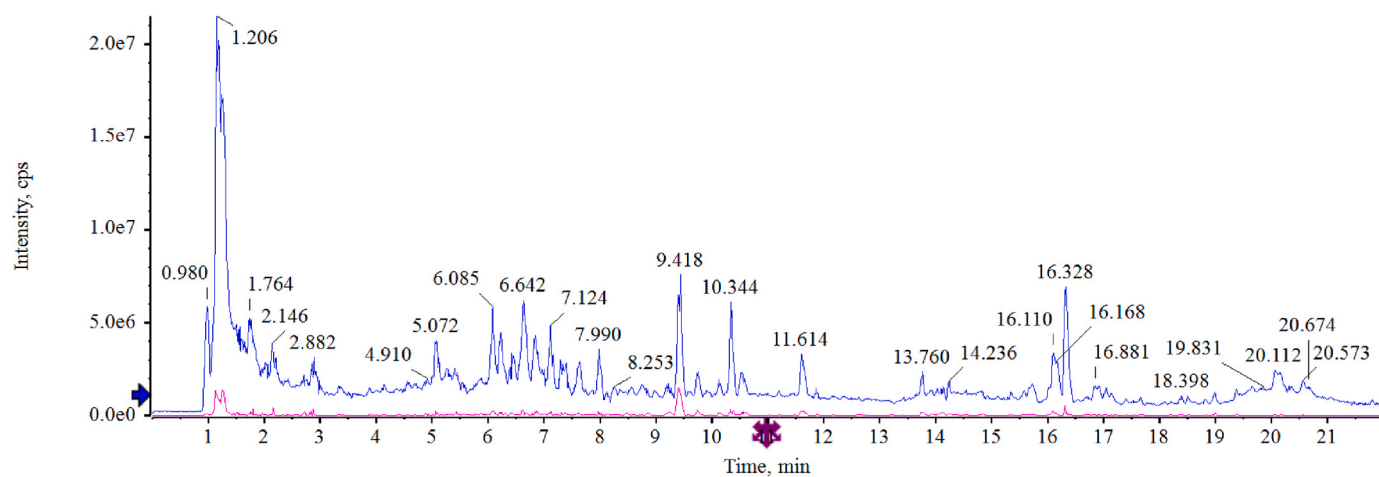
3.10. Ultrastructure of the testicular tissue

Transmission electron microscopy showed that there were large numbers of sperm cells in the NC group with normal shape and structure and an ordered arrangement. The basement membrane was regular, and the Sertoli cells were tightly bound to it. The nuclei were elliptical and obvious, and the mitochondria were abundant in primary spermatocytes. In the SD group, the gap between the spermatogonia and the Sertoli cells was increased, and there were vacuoles. The size and number of lysosomes was increased, and the numbers of primary

Table 6
KEGG pathway enrichment analysis of common targets.

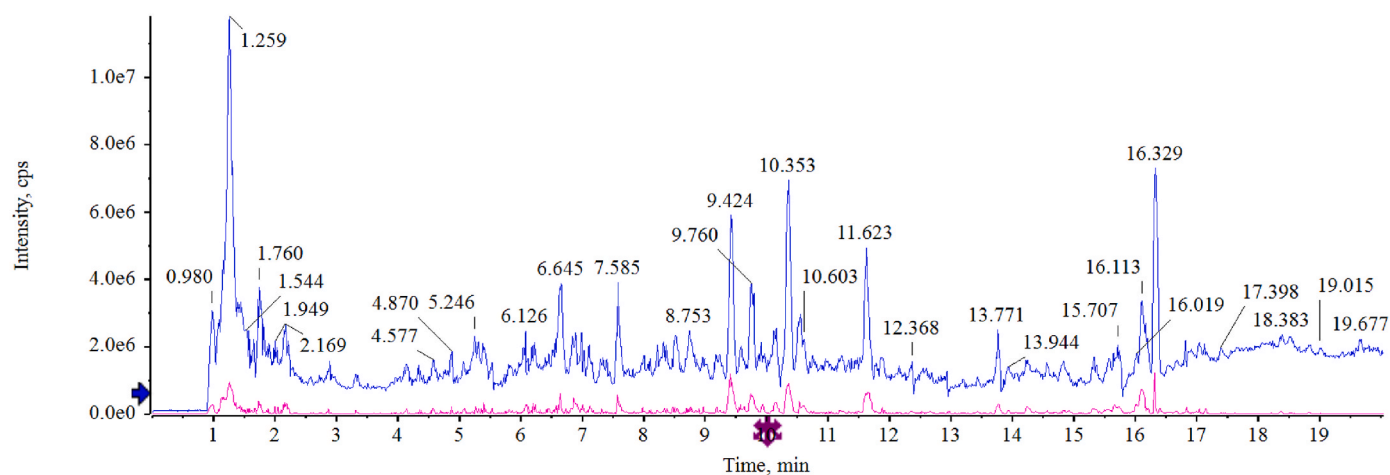
No.	ID	Description	GeneRatio	BgRatio	pvalue	p. adjust	qvalue	geneID	Count
1	hsa05167	Kaposi sarcoma-associated herpesvirus infection	19/43	186/ 8018	1.71E- 20	1.1E- 18	3.19E- 19	5743/595/5970/581/842/3725/836/841/207/ 5599/6772/3569/4609/7422/2353/1026/ 5594/7157/3576	19
2	hsa05161	Hepatitis B	18/43	162/ 8018	4.76E- 20	2.3E- 18	6.64E- 19	5970/581/842/3725/836/841/7040/207/ 5599/6772/3569/4609/2353/1026/5594/ 7157/3576/332	18
3	hsa05163	Human cytomegalovirus infection	17/43	225/ 8018	5.01E- 16	1.07E- 14	3.11E- 15	5743/595/5970/581/842/836/841/207/3569/ 4609/1956/7422/1026/5594/7157/3553/3576	17
4	hsa05210	Colorectal cancer	16/43	86/ 8018	1.49E- 21	2.88E- 19	8.33E- 20	595/581/842/3725/836/7040/207/5599/ 4609/1956/2353/1026/5594/1950/7157/332	16
5	hsa05162	Measles	16/43	138/ 8018	4.37E- 18	1.41E- 16	4.07E- 17	595/5970/581/842/3725/836/841/207/5599/ 6772/3569/598/2353/7157/3553/3558	16
6	hsa05160	Hepatitis C	16/43	155/ 8018	2.94E- 17	8.11E- 16	2.35E- 16	595/5970/581/842/836/841/207/6772/4609/ 1956/1026/5594/1950/7157/3458/5465	16
7	hsa05166	Human T-cell leukemia virus 1 infection	16/43	219/ 8018	7.67E- 15	9.87E- 14	2.85E- 14	595/5970/581/3725/7040/207/5599/3569/ 4609/598/2353/1026/5594/7157/3558/5728	16
8	hsa05212	Pancreatic cancer	15/43	76/ 8018	1.28E- 20	1.1E- 18	3.19E- 19	595/5970/581/842/7040/207/5599/6772/ 1956/7422/598/1026/5594/1950/7157	15
9	hsa04933	AGE-RAGE signaling pathway in diabetic complications	15/43	100/ 8018	1.06E- 18	4.08E- 17	1.18E- 17	595/5970/581/3725/836/7040/207/5599/ 6772/4313/3569/7422/5594/3553/3576	15
10	hsa05225	Hepatocellular carcinoma	15/43	168/ 8018	3.2E- 15	6.18E- 14	1.79E- 14	595/581/7040/207/3162/2950/4609/1956/ 598/1026/5594/7157/5728/4780/1728	15
11	hsa05165	Human papillomavirus infection	15/43	330/ 8018	6.39E- 11	2.74E- 10	7.93E- 11	5743/595/5970/581/836/841/207/6772/ 1956/7422/1026/5594/1950/7157/5728	15
12	hsa04151	PI3K-Akt signaling pathway	15/43	354/ 8018	1.73E- 10	6.94E- 10	2.01E- 10	595/5970/842/207/3569/4609/1956/7422/ 598/1026/5594/1950/7157/3558/5728	15
13	hsa05142	Chagas disease (American trypanosomiasis)	14/43	102/ 8018	6.71E- 17	1.62E- 15	4.68E- 16	3586/5970/3725/841/7040/207/5599/3569/ 2353/5594/3553/3576/3558/3458	14
14	hsa04210	Apoptosis	14/43	136/ 8018	4.27E- 15	7.49E- 14	2.17E- 14	5970/581/842/3725/836/841/207/5599/598/ 2353/5594/7157/332/142	14
15	hsa05418	Fluid shear stress and atherosclerosis	14/43	139/ 8018	5.83E- 15	8.65E- 14	2.5E- 14	5970/3725/207/5599/3162/2950/4313/7422/ 2353/7157/3553/3458/4780/1728	14
16	hsa05152	Tuberculosis	14/43	180/ 8018	2.21E- 13	1.94E- 12	5.61E- 13	3586/5970/581/842/836/841/7040/207/ 5599/6772/3569/5594/3553/3458	14
17	hsa05169	Epstein-Barr virus infection	14/43	201/ 8018	1.02E- 12	7.86E- 12	2.27E- 12	595/5970/581/842/3725/836/841/207/5599/ 6772/3569/4609/1026/7157	14
18	hsa04010	MAPK signaling pathway	14/43	295/ 8018	1.84E- 10	7.22E- 10	2.09E- 10	5970/3725/836/7040/207/5599/4609/1956/ 7422/2353/5594/1950/7157/3553	14
19	hsa05224	Breast cancer	13/43	147/ 8018	3.53E- 13	2.84E- 12	8.21E- 13	595/5241/581/3725/207/4609/1956/2353/ 1026/5594/1950/7157/5728	13
20	hsa04218	Cellular senescence	13/43	160/ 8018	1.06E- 12	7.89E- 12	2.28E- 12	595/5970/7040/207/983/3569/4609/1026/ 5594/7157/3576/891/5728	13
21	hsa05170	Human immunodeficiency virus 1 infection	13/43	212/ 8018	3.89E- 11	1.88E- 10	5.42E- 11	5970/581/842/3725/836/841/207/5599/983/ 598/2353/5594/891	13
22	hsa05132	Salmonella infection	13/43	214/ 8018	4.38E- 11	2E-10	5.79E- 11	5970/581/3725/836/841/207/5599/3569/ 4609/2353/5594/3553/3576	13

- IDA Survey from Tu si zi and gou qi zi 100mL POS.wiff2 (s...(100mL)_POS_Final, Experiment 1, +IDA TOF MS (50 - 1500)
 ● IDA Dependent Sum from Tu si zi and gou qi zi 100mL POS.w...L)_POS_Final, Experiment 2-13, +IDA TOF MSMS (25 - 1500)



(A)

- IDA Survey from Tu si zi and gou qi zi 100mL NEG.wiff2 (... (100mL)_NEG_Final, Experiment 1, -IDA TOF MS (50 - 1500)
 ● IDA Dependent Sum from Tu si zi and gou qi zi 100mL NEG.w...L)_NEG_Final, Experiment 2-13, -IDA TOF MSMS (25 - 1500)



(B)

Fig. 7. The total ion flow diagram of SC-FL by UHPLC-Q/TOF-MS. (A) Positive ion mode; (B) negative ion mode.

Table 7

High resolution mass spectrometry data and elemental composition of SC-FL.

No.	Component Name	Area	Retention Time	Formula	Precursor Mass	Mass Error (ppm)
1	Betaine	2.94 E+07	1.17	C5H11NO2	118.086	-2.9
2	Pipecolic acid	2.04 E+05	1.32	C6H11NO2	130.086	0.8
3	Nicotinic acid	2.59 E+05	1.74	C6H5NO2	124.039	1.3
4	Physcion-8-O-beta-D-gentiobioside	3.18 E+05	9.04	C28H32O15	607.167	-0.4
5	Nicotinamide	2.38 E+05	1.83	C6H6N2O	123.055	1
6	Adenosine	1.89 E+06	2.71	C10H13N5O4	268.104	-0.5
7	6-Hydroxypurine	4.77 E+04	2.93	C5H4N4O	137.046	-0.4
8	Phenylalanine	4.78 E+05	3.88	C9H11NO2	166.086	0.3
9	Chlorogenic acid	1.45 E+06	6.66	C16H18O9	355.102	1
10	Isoscapoletin	6.61 E+05	9.98	C10H8O4	193.05	-0.1
11	Calycosin-7-O-glucoside	2.80 E+04	10.22	C22H22O10	447.129	0.6
12	Quercetin	9.20 E+06	10.34	C15H10O7	303.05	0
13	Berberrubine	4.74 E+04	10.97	C19H15NO4	322.107	1
14	Isorhamnetin	1.36 E+06	11.67	C16H12O7	317.066	0.8
15	Protopine	4.61 E+04	11.78	C20H19NO5	354.134	2
16	4'-O-β-Glucopyranosyl-5-O-methylvisaminol	3.45 E+04	11.8	C22H28O10	453.176	1.5
17	Hesperetin	5.34 E+04	12.35	C16H14O6	303.086	2.1
18	Baicalin	2.14 E+05	13.76	C15H10O5	271.06	1.8
19	Sesamin	3.26 E+03	15.69	C20H18O6	298.364	1.8
20	Ononin	2.24 E+04	13.77	C22H22O9	431.134	2.6
21	Oxypeucedanin hydrate	3.03 E+04	14.38	C16H16O6	305.102	1.3
22	Psoralen	4.81 E+04	15.14	C11H6O3	187.039	1.1
23	Paeonol	5.98 E+05	15.8	C9H10O3	167.07	-0.3
24	Bergapten	1.80 E+04	16.05	C12H8O4	217.05	1.2
25	Formononetin	1.51 E+05	16.06	C16H12O4	269.081	0.6
26	Histidine	7.84 E+04	1.09	C6H9N3O2	154.062	-0.2
27	Serine	6.43 E+04	1.12	C3H7NO3	104.035	0.4
28	D-(+)-Mannose	2.90 E+06	1.2	C6H12O6	179.056	0.2
29	Quinic acid	2.24 E+06	1.23	C7H12O6	191.056	0.5
30	Maleic acid	9.25 E+05	1.34	C4H4O4	115.004	0.1
31	Citric acid	9.50 E+06	1.79	C6H8O7	191.02	1.6
32	Succinic acid	6.89 E+04	2.46	C4H6O4	117.019	0.5
33	Guanosine	1.36 E+05	2.91	C10H13N5O5	282.084	1.4
34	Gallic acid	2.27 E+04	2.97	C7H6O5	169.014	1
35	4-Hydroxybenzoic acid	1.22 E+05	3.61	C7H6O3	137.024	0.8
36	p-Coumaric acid	2.04 E+06	5.4	C9H8O3	163.04	0.9
37	Mandenol	7.55 E+02	21.18	C20H36O2	309.279	3.8
38	L-Tryptophan	2.98 E+05	5.59	C11H12N2O2	203.083	1
39	Isoferulic acid	5.48 E+05	6.17	C10H10O4	193.051	0.6
40	Esculetin	5.78 E+04	7.15	C9H6O4	177.019	0.3
41	Caffeic acid	3.75 E+05	7.33	C9H8O4	179.035	0.5
42	Puerarin	2.05 E+04	7.76	C21H20O9	415.103	-0.1
43	Hyperin	1.96 E+07	10.35	C21H20O12	463.088	0.3
44	IsoActeoside	8.78 E+04	10.59	C29H36O15	623.198	-0.8
45	Cistanoside D + HCOOH	5.92 E+04	11.26	C31H40O15.HCOOH	697.235	-0.4
46	Pinoresinol-glucoside	6.92 E+05	11.63	C26H32O11	519.187	-0.1
47	3,5-Dicaffeoyl quinic acid	6.22 E+05	12.37	C25H24O12	515.119	0.9
48	Naringenin	2.92 E+04	15.18	C15H12O5	271.061	-0.9
49	Chrysosplenetin B	1.10 E+05	16.82	C19H18O8	373.093	-0.8
50	atropine	2.24 E+03	10.67	C17H23NO3	290.175	1.2
51	Ethyl linolenate	1.08 E+03	19.81	C20H34O2	307.263	-0.1
52	cyanin	1.29 E+06	10.15	C27H31O16+	610.154	-6.2
53	kaempferol	5.74 E+06	11.61	C15H10O6	287.055	0
54	Cuscutoside B	4.97 E+05	14.8	C31H36O16	663.193	0.3
55	Lantadene A	2.60 E+04	21.04	C35H52O5	553.389	2.1
57	matrine	5.32 E+0.3	9.84	C15H24N2O	312.025	0.2

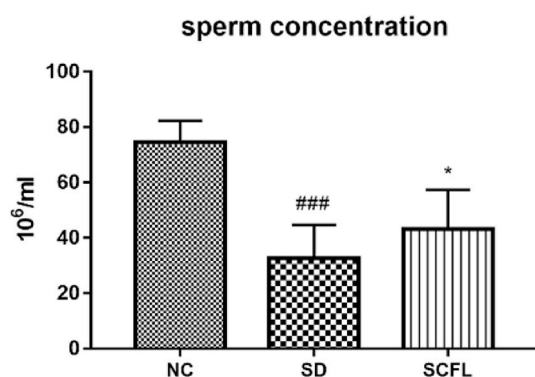


Fig. 8. Sperm concentration of rats from the NC, SD, and SCFL groups. ###P < 0.001 vs. the NC group; *P < 0.05 vs. the SD group.

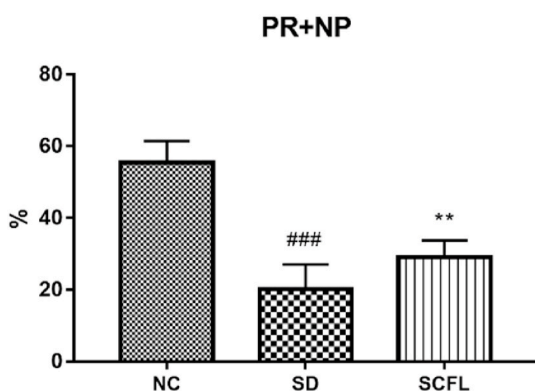


Fig. 9. PR + NP% of rats from the NC, SD, and SCFL groups. ###P < 0.001 vs. the NC group; **P < 0.01 vs. the SD group.

spermatocytes and secondary spermatocytes were decreased. The cells and nuclei were shrunk, the cytoplasm swollen, and the vacuoles had degenerated. There were few sperm particles. A large number of mitochondria could be seen in the primary spermatocytes of rats in the SCFL group. The mitochondrial structure was almost normal, the tubular ridge structures were visible, and there were few vacuolar mitochondria. The structure of primary spermatocytes and secondary spermatocytes was improved. Sperm particles could be seen in the secondary spermatocytes, and the number of sperm cells was increased. Structural damage was reduced to a certain extent, but the cells were small, and there was still some vacuolar degeneration. The specific results are shown in Fig. 11.

3.11. Protein expression of *c-jun*, *p38 MAPK*, *Occludin*, *ZO-1*, β -catenin, *N-cadherin* and *connexin 43* in the testis

The protein electrophoretograms of *c-jun*, *p38 MAPK*, *Occludin*, *ZO-1*, β -catenin, *N-cadherin* and *connexin 43* in the testicular tissues of the three groups are shown in Fig. 11. Compared with the NC group, the protein expression of *c-jun*, *Occludin*, *ZO-1*, β -catenin and *connexin 43* in testicular tissue of the SD group was significantly lower, and the differences were statistically significant ($P < 0.05$ or $P < 0.01$). The protein expression of *p38 MAPK* in the SD group increased significantly, and the differences were statistically significant ($P < 0.05$). *N-cadherin* showed no statistically significant difference but showed a certain trend toward reduction. Compared with the rats in the SD group, the protein expression of *c-jun*, *Occludin*, *ZO-1* and *connexin 43* in the SCFL group was significantly increased, and *p38 MAPK* expression was significantly reduced. Compared with the SD group, β -catenin and *N-cadherin* expression in the SCFL group showed no statistically significant

differences but showed a certain trend. The specific results are shown in Fig. 12.

3.12. mRNA expression of *c-jun*, *p38 MAPK*, *Occludin*, *ZO-1*, β -catenin, *N-cadherin* and *connexin 43* in the testis

Compared with those in the NC group, the mRNA expression of *c-jun*, *Occludin*, *ZO-1*, β -catenin and *connexin 43* in testicular tissue in the SD group was significantly lower, and the differences were statistically significant ($P < 0.05$ or $P < 0.01$). The mRNA expression of *p38 MAPK* in the SD group increased significantly ($P < 0.05$). *N-cadherin* showed no significant difference but showed a certain trend toward reduction. Compared with those in the rats in the SD group, the mRNA expression of *c-jun*, *Occludin*, *ZO-1* and *connexin 43* in the SCFL group were significantly higher, and the mRNA expression of *p38 MAPK* was significantly lower. Compared with those in the SD group, β -catenin and *N-cadherin* expression in the SCFL group showed no statistically significant differences but showed a certain trend. The specific results are shown in Fig. 13.

4. Discussion

In TCM theory, “the kidney stores essence and governs the reproductive system” has always been a summary of the relationship between the kidney and reproductive function. Sperm quality is believed to depend on the kidney’s essence, and poor semen quality is due to kidney deficiency (Zheng et al., 2017). Western medicine has a better effect in male infertility with a clear cause, however, we cannot find specific cause in a large number of male infertility patients. TCM are widely used and exert curative effects in male infertility. Wuzi Yanzong pills, Zuogui pills, and Yougui pills are all classic prescriptions and show a curative effect on male infertility. A formula often contains several herbs, and each herb has many components, which make formula study difficult. The herb pair consist of two herbs and has fewer components, and it can indicate the regularity of TCM prescriptions and the connotation of syndrome differentiation (Wang et al., 2012). Semen Cuscutae and Fructus Lycii are both in the classic formulae above and widely used for treating male infertility (Soni et al., 2018; Wang et al., 2002). Semen Cuscutae and Fructus Lycii have become a usual herb pair “SC-FL”, and it embodies the treatment concept of male infertility “tonifying the kidney and replenishing the essence”. SC-FL is an effective herb pair for the treatment of male infertility, but its treatment mechanism has not been fully elucidated.

GTW is a fat-soluble mixture extracted and refined from *Tripterygium wilfordii* roots (Xu et al., 2016). It is often used as an immunosuppressant in clinical practice. A large number of studies have shown that GTW has a significant anti-fertility effect (Guo et al., 2019), and GTW-induced spermatogenic dysfunction in a rat model of semen quality in terms of sperm concentration, mobility, and other properties is highly consistent with the pathological characteristics of male infertility (Guan et al., 2020). Moreover, this method has the advantages of easy operation, high specificity and a short modeling time, and it has been widely used to establish spermatogenesis dysfunction.

In this study, we screened 61 active ingredients including quercetin, isorhamnetin, etc. through UPLC-Q/TOF-MS analysis. The above ingredients were sought in TCMSP database.

In this study, we searched the TCMSP database, and 213 ingredients of SC-FL were found, and 54 biologically active ingredients were finally obtained by the screening criteria of OB and DL. At the same time, we performed mass spectrometry analysis. We found that the above results were credible when integrating the results of mass spectrometry analysis and article reference. Then, we further predicted the common targets of SC-FL, GTW, and male infertility, and obtained 44 targets. *C-jun*, *IL6*, *MAPK1*, *TP53*, *MYC*, *CCND1*, *AR*, *EGF*, *FOS* and *MAPK8* were identified as the core targets through the MCC algorithm. We found that SC-FL treatment of male infertility may be related to cellular processes

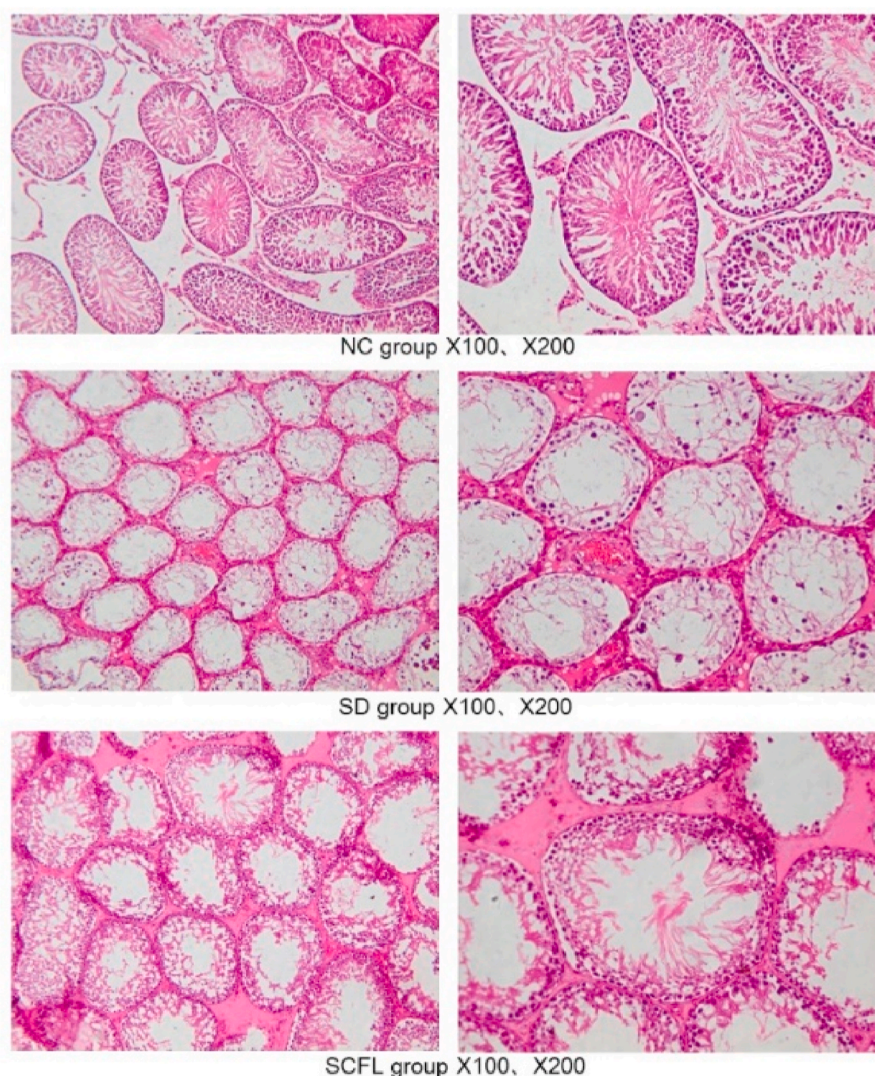


Fig. 10. H&E staining of testicular tissues of rats from the NC, SD, and SCFL groups.

involving active oxygen, oxidative stress, cell proliferation, apoptosis, and signaling pathways such as the PI3K-Akt, apoptosis, MAPK, and cellular senescence pathways through GO enrichment and KEGG pathway enrichment analyses of common targets. To a certain extent, these findings reveal the involvement of multiple targets and systemic mechanisms of SC-FL in treating male infertility.

Spermatogenesis is a very complex process that depends on the coordinated and orderly regulation of multiple systems, such as the endocrine system, immune system, and BTB (Neto et al., 2016). Any abnormality in any step can affect spermatogenesis. The BTB is formed by the interaction of Sertoli cells and surrounding structures. The BTB can regulate the nutrients and important molecules essential for spermatogenesis and resist harmful poisons entering the lumen of seminiferous tubules to keep the microenvironment of spermatogenesis stable (Griswold, 1998; Mruk and Cheng, 2015). Sertoli cells not only directly participate in the formation of the BTB and play a structural supporting role but also secrete INHB, FSH, regulatory proteins, growth factors, and other molecules to provide energy and nutrition for spermatogenesis. The BTB is composed of tight junctions (TJs), adhesive junctions (AJs), and gap junctions (GJs) (Mruk and Cheng, 2015), which depend on the maintenance of the three junction structures and the coordination of orderly opening and closing (Cheng and Mruk, 2012). TJs are junction structures formed by adjacent Sertoli cells, and the expressed proteins include Occludin and ZO-1. AJs exist between Sertoli cells and between

Sertoli and germ cells, and their related proteins include β -catenin and N-cadherin (Mruk and Cheng, 2004). GJs are formed by the junction of protein semichannels of adjacent cell membranes. GJs are a chemical pathway between Sertoli cells and between Sertoli and spermatogenic cells. Connexin 43 is the most abundantly expressed protein in GJs (Griswold, 2018). Therefore, morphological and functional abnormalities of Sertoli cells or abnormal expression of BTB junction structure-related proteins can reflect BTB damage. The disorder of BTB structure and function or the abnormal expression of related proteins will affect the proliferation and differentiation of spermatogenic cells.

Several studies have shown that GTW changes the function of Sertoli cells (Lan et al., 1992; Li et al., 2009; Xiong et al., 2011), disrupts the integrity of the BTB and leads to a decline in spermatogenic function. Our previous research has confirmed that GTW can promote the apoptosis of spermatogenic cells by regulating the PI3K/Akt pathway, and SC-FL can reduce the destructive effect of GTW (Guan et al., 2020), which also confirms the credibility of the results of network pharmacology prediction in this study. We found that the reproductive toxicity of GTW not only directly affects the apoptosis of spermatogenic cells but also leads to spermatogenic dysfunction by disrupting the BTB. Its role is related to the MAPK signaling pathway. Therefore, the spermatogenic dysfunction model constructed with GTW can well simulate the infertility caused by mitochondrial apoptosis and BTB damage, and the SC-FL has a good therapeutic effect on this model.

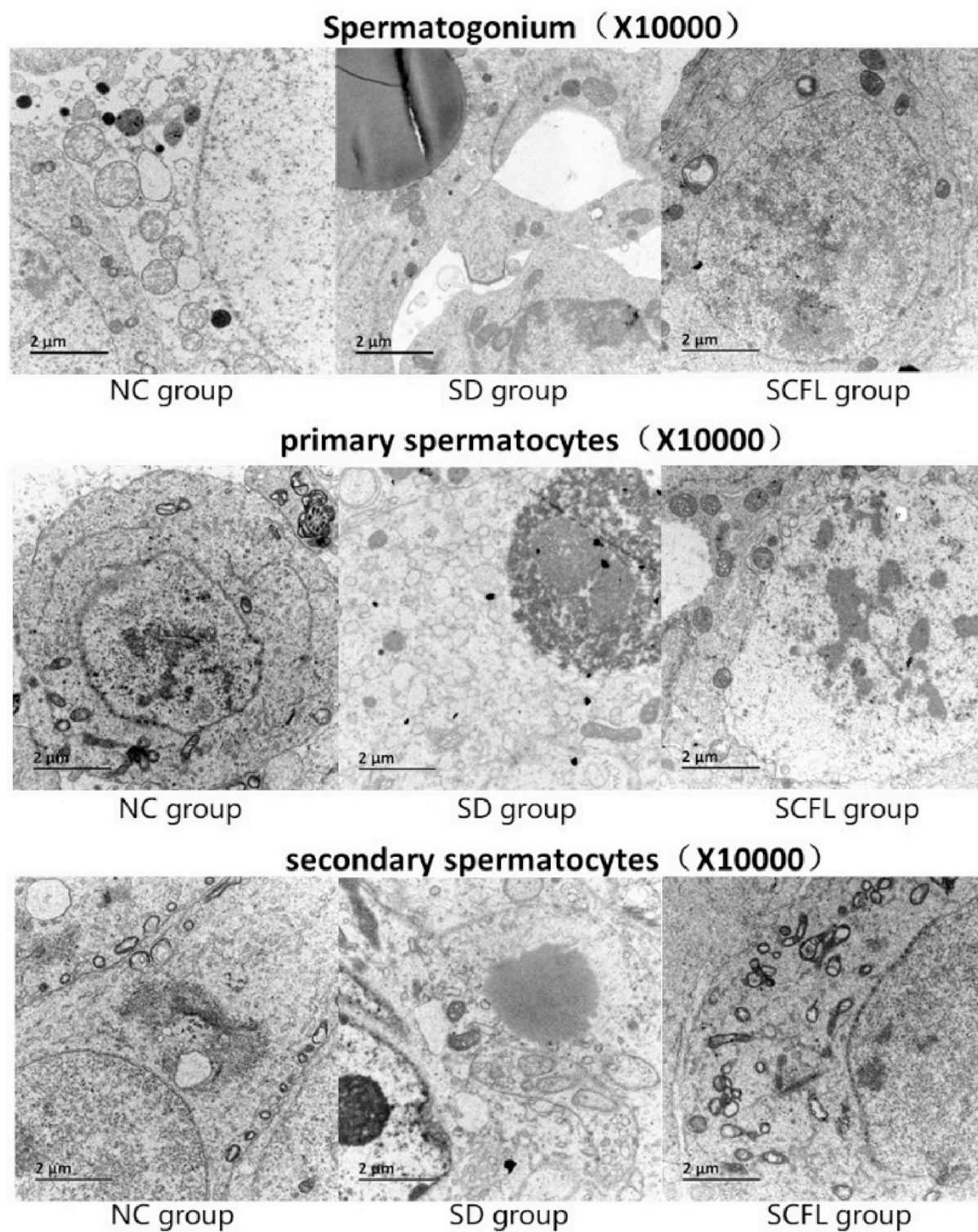


Fig. 11. Ultrastructure of testicular tissue of rats from the NC, SD, and SCFL groups (x10000).

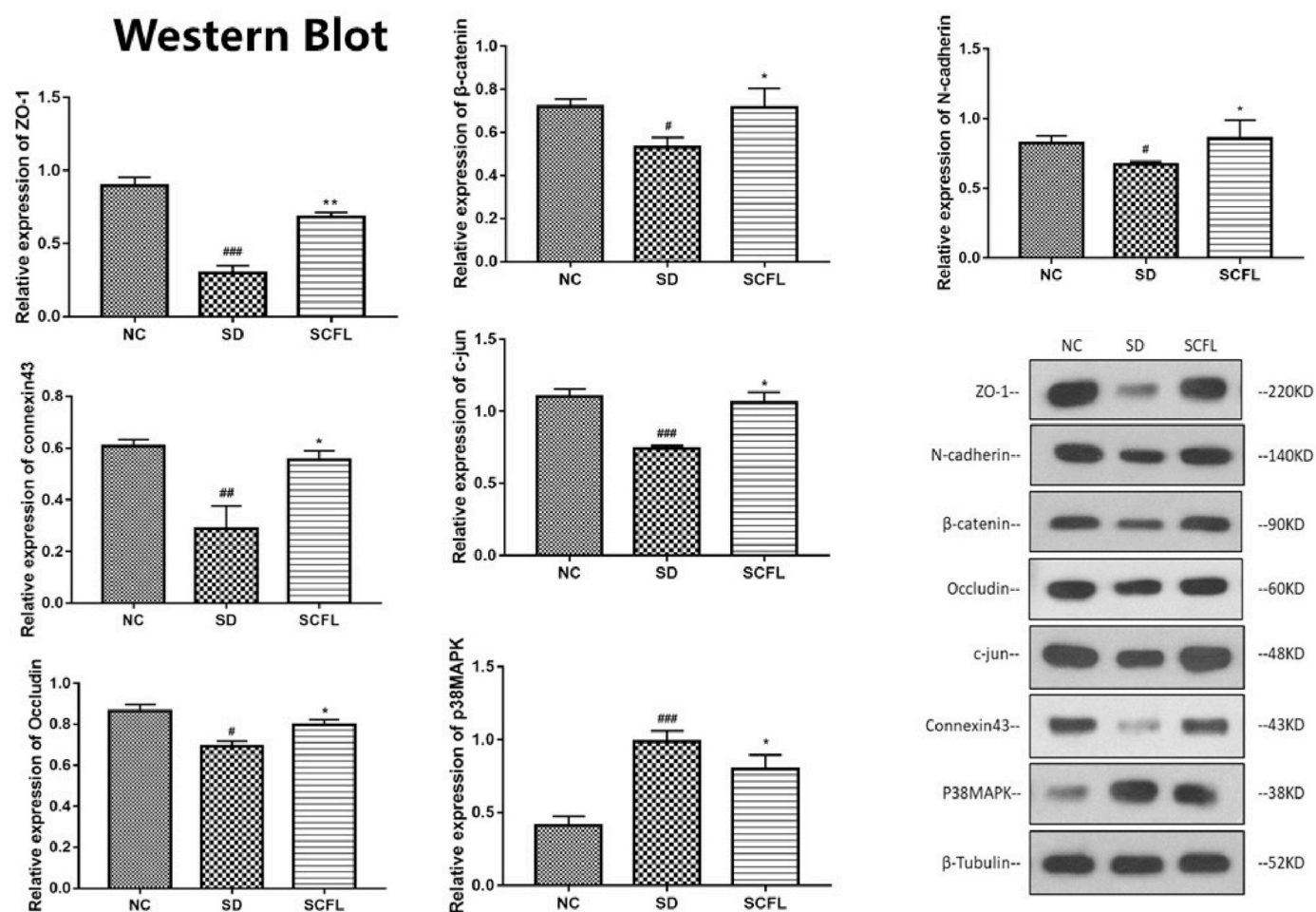


Fig. 12. Electrophoretogram and expression levels of c-jun, p38MAPK, Occludin, ZO-1, β-catenin, N-cadherin and connexin43 in rats from the NC, SD, and SCFL groups determined using western blotting. # $P < 0.05$, ## $P < 0.01$ and ### $P < 0.001$ vs. the NC group; * $P < 0.05$ and ** $P < 0.01$ vs. the SD group.

The mitogen-activated protein kinase (MAPK) signaling pathway plays an important role in the destruction of the BTB structure and function induced by exogenous poisons (Liu et al., 2020; Qiu et al., 2016). The most important members of the MAPK pathway include p38 mitogen-activated protein kinase (p38 MAPK), extracellular signal-regulated kinase (ERK) and Jun N-terminal kinase (JNK) (Qiu et al., 2013). The P38 signaling pathway is a more extensively studied branch of the MAPK signaling pathway. Many studies have proven that the p38 MAPK pathway can regulate the expression of BTB proteins such as Claudin-11, Connexin 43 and Occludin in the testis. When these proteins are expressed at low levels through the p38 MAPK pathway, they can cause male reproductive function damage (Lu et al., 2016). At the same time, under hypoxic conditions, the expression of Occludin (a TJ protein) is reduced through this signaling pathway, causing cytoskeletal destruction and cell apoptosis (Wagner and Nebreda, 2009). JNK is also an important protein that affects the stability of the BTB. Studies have shown that the addition of JNK pathway inhibitors can block BTB destruction induced by poison (Chen et al., 2018; Ma et al., 2019).

By observing the HE-stained sections and the ultrastructure of the cells under the electron microscope, we proved that SC-FL can improve the pathological state in the testis and repair the damaged seminiferous tubules. SC-FL inhibits the apoptosis of Sertoli cells and spermatogenic cells, and it mainly acts on mitochondria. In terms of BTB junction structure-related proteins, compared with that in the SD group, p38 MAPK expression in the SC-FL group decreased significantly, and the protein expression of c-Jun, Occludin, ZO-1, and connexin 43

significantly increased, indicating that SC-FL can regulate the secretion function of Sertoli cells and BTB junction structure-related proteins, thereby maintaining the integrity of the BTB to reduce spermatogenic dysfunction. The results of mechanism of signal pathway showed that SC-FL can significantly up-regulate the expression of protein and mRNA of p38MAPK and c-jun, indicating that the mechanism of SC-FL may be related to the p38 pathway and JUN pathway in MAPK pathway.

5. Conclusion

In summary, we predicted that the active ingredients of SC-FL can treat male infertility by interacting with the core targets JUN, IL6, MAPK1, TP53, MYC, CCND1, AR, EGF, FOS, and MAPK8, and the possible mechanism is related to the MAPK signaling pathway. Animal experiments helped to confirm that SC-FL has an obvious therapeutic effect on spermatogenic dysfunction induced by GTW. SC-FL can regulate the MAPK signaling pathway and affect the expression of Occludin, ZO-1 and connexin 43 to repair damaged BTB and improve spermatogenic function, which may be one of the possible mechanisms. However, the BTB is only one component associated with spermatogenesis, and there are many related mechanisms and pathways involved in spermatogenesis. Whether SC-FL affects other pathways and expression of other proteins has yet to be further explored and studied.

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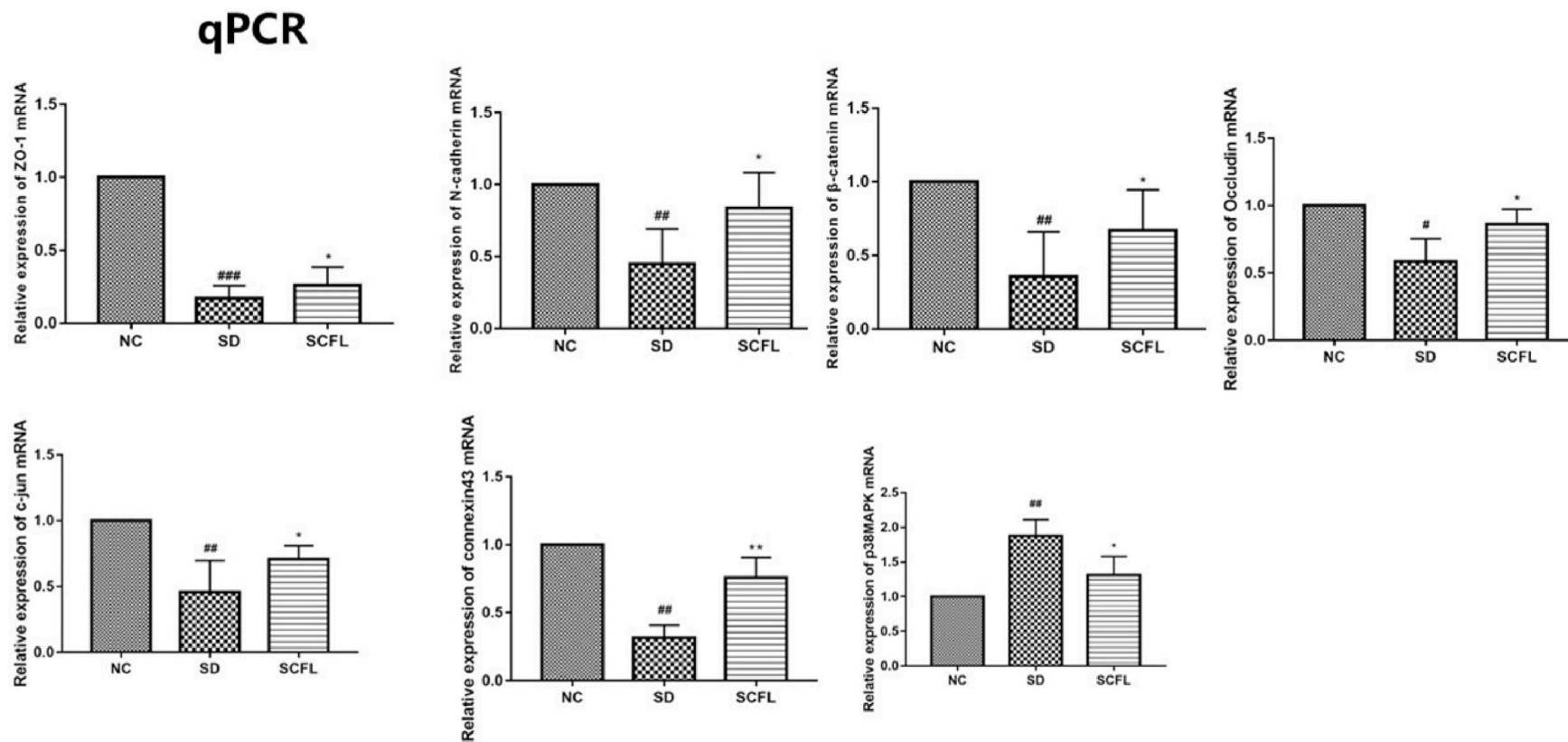


Fig. 13. The mRNA expression levels of c-jun, p38MAPK, Occludin, ZO-1, β-catenin, N-cadherin and connexin43 in rats from the NC, SD, and SCFL groups quantified using PCR. #P < 0.05, ##P < 0.01 and ###P < 0.001 vs. the NC group; *P < 0.05 and **P < 0.01 vs. the SD group.

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Authors' contributions

YZ, LD, and JW performed the investigation, analyzed the data and wrote the manuscript; YZ, YL and SG designed the study and amended the manuscript; LD, JW, QL, and HT helped in the execution of the research; SG, YZ, LD, JW, and YL read, improved and approved the manuscript.

Declaration of competing interest

The authors declare that there is no conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2021.114022>.

References

- Amberger, J.S., Bocchini, C.A., Schiettecatte, F., Scott, A.F., Hamosh, A., 2015. OMIM.org: online Mendelian Inheritance in Man (OMIM®), an online catalog of human genes and genetic disorders. *Nucleic Acids Res.* 43, D789–D798.
- Chen, Y., Wang, J., Pan, C., Li, D., Han, X., 2018. Microcystin-leucine-arginine causes blood-testis barrier disruption and degradation of occludin mediated by matrix metalloproteinase-8. *Cell. Mol. Life Sci.* : CM 75, 1117–1132.
- Cheng, C.Y., Mruk, D.D., 2012. The blood-testis barrier and its implications for male contraception. *Pharmacol. Rev.* 64, 16–64.
- Davis, A.P., Grondin, C.J., Johnson, R.J., Sciaky, D., McMoran, R., Wiegiers, J., Wiegiers, T.C., Mattingly, C.J., 2019. The comparative Toxicogenomics database: update 2019. *Nucleic Acids Res.* 47, D948–d954.
- Du, X., He, X., Huang, Y., Fu, B., Liang, B., Lv, C., Li, Y., Gao, X., Wang, Y., 2019. Simultaneous determination of seven effective components of Tripterygium glycosides in human biological matrices by ultra performance liquid chromatography-triple quadrupole mass spectrometry. *Journal of chromatography B, Analytical technologies in the biomedical and life sciences* 1113, 1–13.
- Du, X., Nyagblordzro, M., An, L., Gao, X., Du, L., Wang, Y., Ondieki, G., Kikete, S., He, X., 2018. Pharmacokinetic and toxicological characteristics of tripterygium glycosides and their derivatives. *Curr. Drug Metabol.* 19, 605–627.
- Griswold, M.D., 2018. 50 years of spermatogenesis: sertoli cells and their interactions with germ cells. *Biol. Reprod.* 99, 87–100.
- Griswold, M.D., 1998. The central role of Sertoli cells in spermatogenesis. *Semin. Cell Dev. Biol.* 9, 411–416.
- Guan, S., Zhu, Y., Wang, J., Dong, L., Zhao, Q., Wang, L., Wang, B., Li, H., 2020. A combination of Semen Cuscutae and Fructus Lycii improves testicular cell proliferation and inhibits their apoptosis in rats with spermatogenic dysfunction by regulating the SCF/c-kit–PI3K–Bcl-2 pathway. *J. Ethnopharmacol.* 251, 112525.
- Guo, J., Huang, Y., Lei, X., Zhang, H., Xiao, B., Han, Z., Liang, C., Yang, W., 2019. Reproductive systemic toxicity and mechanism of glucosides of Tripterygium wilfordii Hook. F. (GTW). *Ann. Clin. Lab. Sci.* 49, 36–49.
- He, X., Taylor, A., Jarvis, M., Moshin, J., Morla, A., 2016. Efficient forensic toxicological screening and quantitation workflow using QTOF LC–MS/MS system. *Toxicologie Analytique Et Clinique* 28, S43–S43.
- Jiang, D., Coscione, A., Li, L., Zeng, B.Y., 2017. Effect of Chinese herbal medicine on male infertility. *Int. Rev. Neurobiol.* 135, 297–311.
- Krausz, C., Riera-Escamilla, A., 2018. Genetics of male infertility. *Nat. Rev. Urol.* 15, 369–384.
- Lan, Z.J., Gu, Z.P., Lu, R.F., Zhuang, L.Z., 1992. Effects of multiglycosides of Tripterygium wilfordii (GTW) on rat fertility and Leydig and Sertoli cells. *Contraception* 45, 249–261.
- Li, J., Huang, D., He, L.Q., 2009. [Effects of glycosides of Tripterygium wilfordii and Cistanche deserticola on the fertility of male mice]. *Zhonghua nan ke xue = National journal of andrology* 15, 569–572.
- Liu, B., Shen, L.J., Zhao, T.X., Sun, M., Wang, J.K., Long, C.L., He, D.W., Lin, T., Wu, S.D., Wei, G.H., 2020. Automobile exhaust-derived PM(2.5) induces blood-testis barrier damage through ROS-MAPK-Nrf2 pathway in sertoli cells of rats. *Ecotoxicol. Environ. Saf.* 189, 110053.
- Lu, Y., Luo, B., Li, J., Dai, J., 2016. Perfluorooctanoic acid disrupts the blood-testis barrier and activates the TNF α /p38 MAPK signaling pathway in vivo and in vitro. *Arch. Toxicol.* 90, 971–983.
- Ma, B., Zhang, J., Zhu, Z., Bao, X., Zhang, M., Ren, C., Zhang, Q., 2019. Aucubin, a natural iridoid glucoside, attenuates oxidative stress-induced testis injury by inhibiting JNK and CHOP activation via Nrf2 up-regulation. *Phytomedicine : international journal of phytotherapy and phytopharmacology* 64, 153057.
- Mruk, D.D., Cheng, C.Y., 2015. The mammalian blood-testis barrier: its biology and regulation. *Endocr. Rev.* 36, 564–591.
- Mruk, D.D., Cheng, C.Y., 2004. Sertoli-Sertoli and Sertoli-germ cell interactions and their significance in germ cell movement in the seminiferous epithelium during spermatogenesis. *Endocr. Rev.* 25, 747–806.
- Neto, F.T., Bach, P.V., Najari, B.B., Li, P.S., Goldstein, M., 2016. Spermatogenesis in humans and its affecting factors. *Semin. Cell Dev. Biol.* 59, 10–26.
- Piñero, J., Bravo, A., Queralt-Rosinach, N., Gutiérrez-Sacristán, A., Deu-Pons, J., Centeno, E., García-García, J., Sanz, F., Furlong, L.I., 2017. DisGenET: a comprehensive platform integrating information on human disease-associated genes and variants. *Nucleic Acids Res.* 45, D833–d839.
- Qiu, L., Qian, Y., Liu, Z., Wang, C., Qu, J., Wang, X., Wang, S., 2016. Perfluorooctane sulfonate (PFOS) disrupts blood-testis barrier by down-regulating junction proteins via p38 MAPK/ATF2/MMP9 signaling pathway. *Toxicology* 373, 1–12.
- Qiu, L., Zhang, X., Zhang, X., Zhang, Y., Gu, J., Chen, M., Zhang, Z., Wang, X., Wang, S.L., 2013. Sertoli cell is a potential target for perfluorooctane sulfonate-induced reproductive dysfunction in male mice. *Toxicol. Sci.: an official journal of the Society of Toxicology* 135, 229–240.
- Ru, J., Li, P., Wang, J., Zhou, W., Li, B., Huang, C., Li, P., Guo, Z., Tao, W., Yang, Y., Xu, X., Li, Y., Wang, Y., Yang, L., 2014. TCMS: a database of systems pharmacology for drug discovery from herbal medicines. *J. Cheminf.* 6, 13.
- Safran, M., Dalah, I., Alexander, J., Rosen, N., Iny Stein, T., Shmoish, M., Nativ, N., Bahir, I., Doniger, T., Krug, H., Sirota-Madi, A., Olender, T., Golani, Y., Stelzer, G., Harel, A., Lancet, D., 2010. GeneCards Version 3: the human gene integrator. *Database : the journal of biological databases and curation* 2010, baq020.
- Soni, K.K., Zhang, L.T., Choi, B.R., Karna, K.K., You, J.H., Shin, Y.S., Lee, S.W., Kim, C.Y., Zhao, C., Chae, H.J., Kim, H.K., Park, J.K., 2018. Protective effect of MOTILPERM in varicocele-induced oxidative injury in rat testis by activating phosphorylated inositol requiring kinase 1 α (p-IRE1 α) and phosphorylated c-Jun N-terminal kinase (p-JNK) pathways. *Pharmaceut. Biol.* 56, 94–103.
- Szklarczyk, D., Gable, A.L., Lyon, D., Junge, A., Wyder, S., Huerta-Cepas, J., Simonovic, M., Doncheva, N.T., Morris, J.H., Bork, P., Jensen, L.J., Mering, C.V., 2019. STRING v11: protein-protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Res.* 47, D607–d613.
- Wagner, E.F., Nebreda, A.R., 2009. Signal integration by JNK and p38 MAPK pathways in cancer development. *Nat. Rev. Canc.* 9, 537–549.
- Wang, S., Hu, Y., Tan, W., Wu, X., Chen, R., Cao, J., Chen, M., Wang, Y., 2012. Compatibility art of traditional Chinese medicine: from the perspective of herb pairs. *J. Ethnopharmacol.* 143, 412–423.
- Wang, X., Shen, Y., Wang, S., Li, S., Zhang, W., Liu, X., Lai, L., Pei, J., Li, H., 2017. PharmMapper 2017 update: a web server for potential drug target identification with a comprehensive target pharmacophore database. *Nucleic Acids Res.* 45, W356–w360.
- Wang, Y., Zhang, S., Li, F., Zhou, Y., Zhang, Y., Wang, Z., Zhang, R., Zhu, J., Ren, Y., Tan, Y., Qin, C., Li, Y., Li, X., Chen, Y., Zhu, F., 2020. Therapeutic target database 2020: enriched resource for facilitating research and early development of targeted therapeutics. *Nucleic Acids Res.* 48, D1031–d1041.
- Wang, Y., Zhao, H., Sheng, X., Gambino, P.E., Costello, B., Bojanowski, K., 2002. Protective effect of Fructus Lycii polysaccharides against time and hyperthermia-induced damage in cultured seminiferous epithelium. *J. Ethnopharmacol.* 82, 169–175.
- Wishart, D.S., Feunang, Y.D., Guo, A.C., Lo, E.J., Marcu, A., Grant, J.R., Sajed, T., Johnson, D., Li, C., Sayeeda, Z., Assempour, N., Iynkkaran, I., Liu, Y., Maciejewski, A., Gale, N., Wilson, A., Chin, L., Cummings, R., Le, D., Pon, A., Knox, C., Wilson, M., 2018. DrugBank 5.0: a major update to the DrugBank database for 2018. *Nucleic Acids Res.* 46, D1074–d1082.
- Xiong, J., Wang, H., Guo, G., Wang, S., He, L., Chen, H., Wu, J., 2011. Male germ cell apoptosis and epigenetic histone modification induced by Tripterygium wilfordii Hook F. *PLoS One* 6, e20751.
- Xu, X., Li, Q.J., Xia, S., Wang, M.M., Ji, W., 2016. Tripterygium glycosides for treating late-onset rheumatoid arthritis: a systematic review and meta-analysis. *Alternative Ther. Health Med.* 22, 32–39.
- Yang, A., Yu, C., You, F., He, C., Li, Z., 2018. Mechanisms of Zuogui pill in treating osteoporosis: perspective from bone marrow mesenchymal stem cells. *Evidence-based complementary and alternative medicine. eCAM* 2018, 3717391.
- Yu, X., Yu, Q., Chang, D., Li, J., Li, Y., Dong, L., 2019. Distribution of traditional Chinese medicine syndrome types of male infertility in Chengdu area. *Zhonghua Nan ke Xue* 25, 909–913.
- Zhao, M., Chan, C.P.S., Cheung, C.W.C., Alqawasmeh, O., Wang, R.C.C., Wu, J.C.Y., Lin, Z.X., Li, T.C., Chung, J.P.W., Mak, J.S.M., Law, T.S.M., Chan, D.Y.L., 2019. A double-blinded, randomized placebo-controlled trial on the effect of traditional Chinese medicine formula Wuzi Yanzong pill on improving semen qualities in men with suboptimal parameters. *Trials* 20, 540.
- Zhang, Z., Yang, J., Kong, T., Wang, C., Ning, P., Chen, X., Li, X., Jia, Y., Chen, X., Liu, S., Zhang, Y., 2018. Traditional Chinese Medicine Syndrome elements of male infertility revealed by latent tree model analysis. *J. Tradit. Chin. Med.* 38, 926–935.
- Zheng, P., Wang, Y., Lu, H., Zhou, X., Tang, T., Fan, R., Zhang, C., Cui, H., Wang, Y., Luo, J., 2017. Plasma metabolomics analysis based on GC-MS in infertile males with kidney-yang deficiency syndrome. *Evidence-based complementary and alternative medicine. eCAM* 6270195, 2017.